

INFORMATION TECHNOLOGY INSTITUTE

Analog IC Design Track



Analog Integrated Circuit Design

Lab 11

Mini Project 02

Fully-Differential Folded Cascode OTA

Table of Contents

1	Part 1: gm/ID Design Charts.....	4
1.1	NMOS.....	4
1.2	PMOS.....	6
2	PART 2: OTA Design.....	8
2.1	Specs:.....	8
2.2	Sizing:.....	9
2.2.1	Input Pair:.....	9
2.2.2	NMOS Current source:.....	9
2.2.3	Cascode NMOS:.....	10
2.2.4	Cascode PMOS:.....	10
2.2.5	Tail PMOS:.....	10
2.2.6	PMOS Current Source:.....	10
3	PART 3: Open-Loop OTA (Behavioral CMFB).....	12
3.1	OP Analysis:.....	12
3.2	Diff small signal ccs:.....	13
4	PART 4: Open-Loop OTA Simulation (Actual CMFB).....	16
4.1	CMFB Design:.....	16
4.2	OP Analysis:.....	17
4.3	Diff small signal ccs:.....	19
5	PART 5: Closed Loop Simulation (AC and STB).....	20
5.1	OP Analysis:.....	21
5.2	Differential closed-loop response:.....	22
5.3	Differential and CMFB loops stability (STB analysis):.....	23
6	PART 6: Closed Loop Simulation (Transient):.....	27
6.1	Differential and CMFB loops stability:.....	27
6.2	Differential and CMFB loops stability (transient analysis):.....	29
6.3	Output swing:.....	29
6.3.1	CMIR Range:.....	31
7	Specs Achieved.....	31
8	Bonus Part:.....	32
8.1	Bias Circuit:.....	32

8.1.1	AC analysis :	34
9	FOM	34
9.1	Total Current	34
9.1.1	OTA	34
9.1.2	CMFB	35
9.2	AREA	35
9.2.1	OTA	35
9.2.2	CMFB	35

1 Part 1: g_m/I_D Design Charts

1.1 NMOS

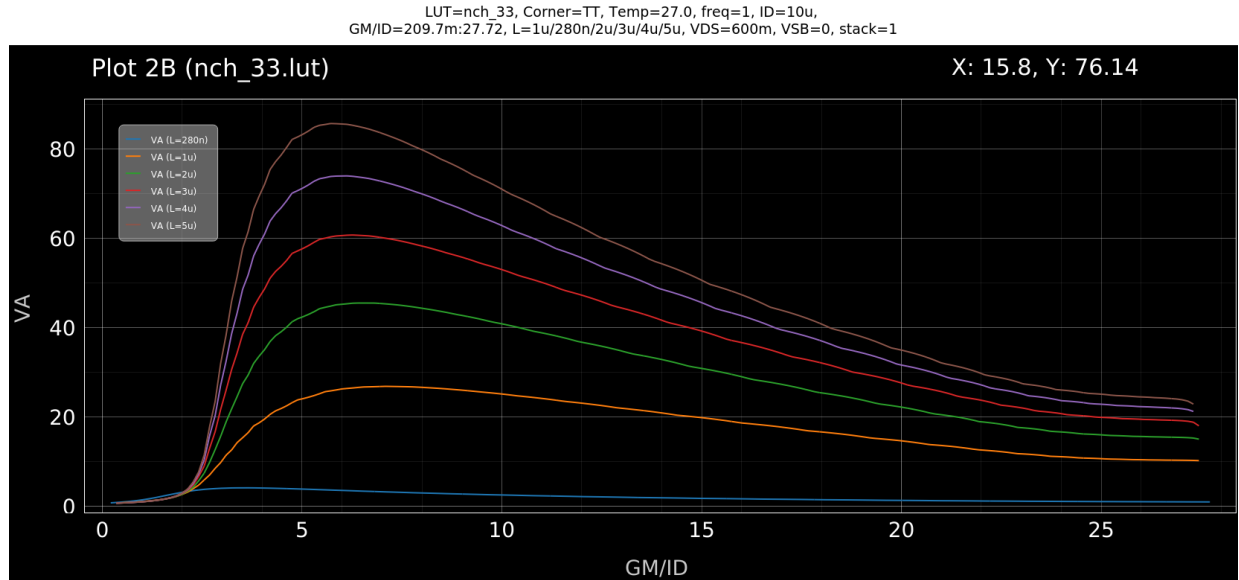


Figure 1: NMOS (V_A) vs. g_m/I_D .

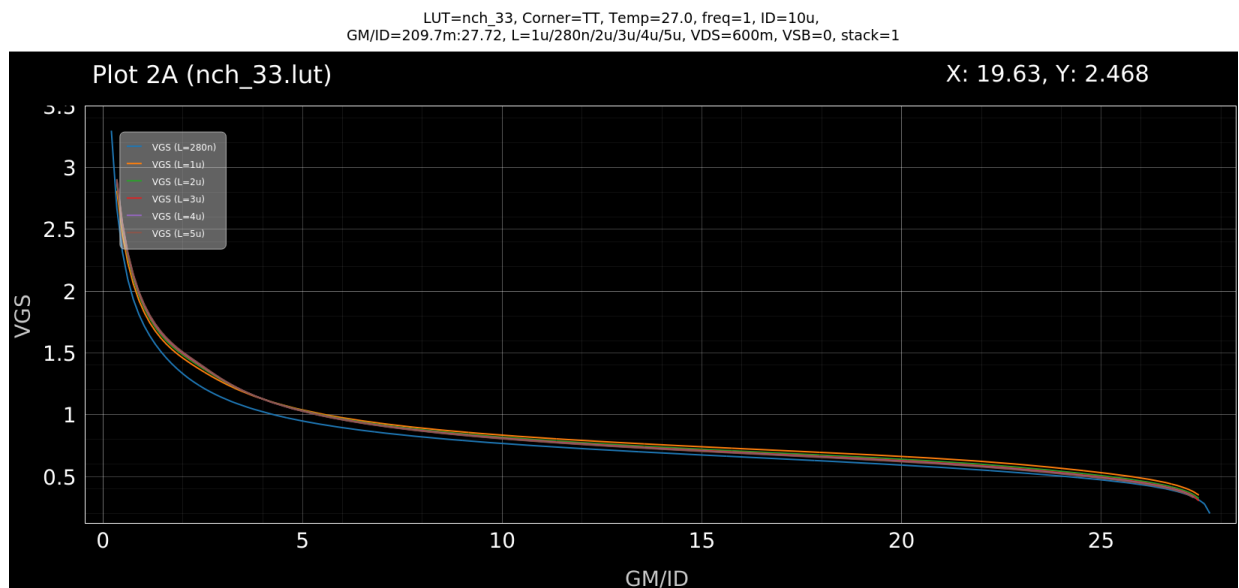


Figure 2: NMOS V_{GS} vs. g_m/I_D .

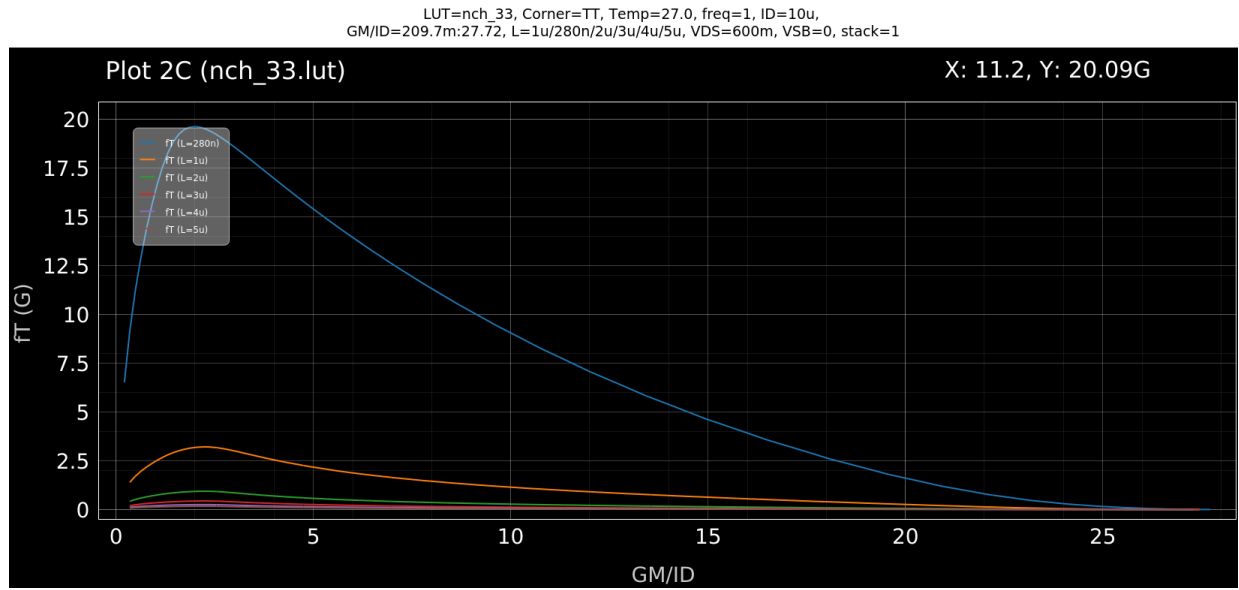


Figure 3: NMOS Transit Frequency (f_T) vs. g_m/I_D

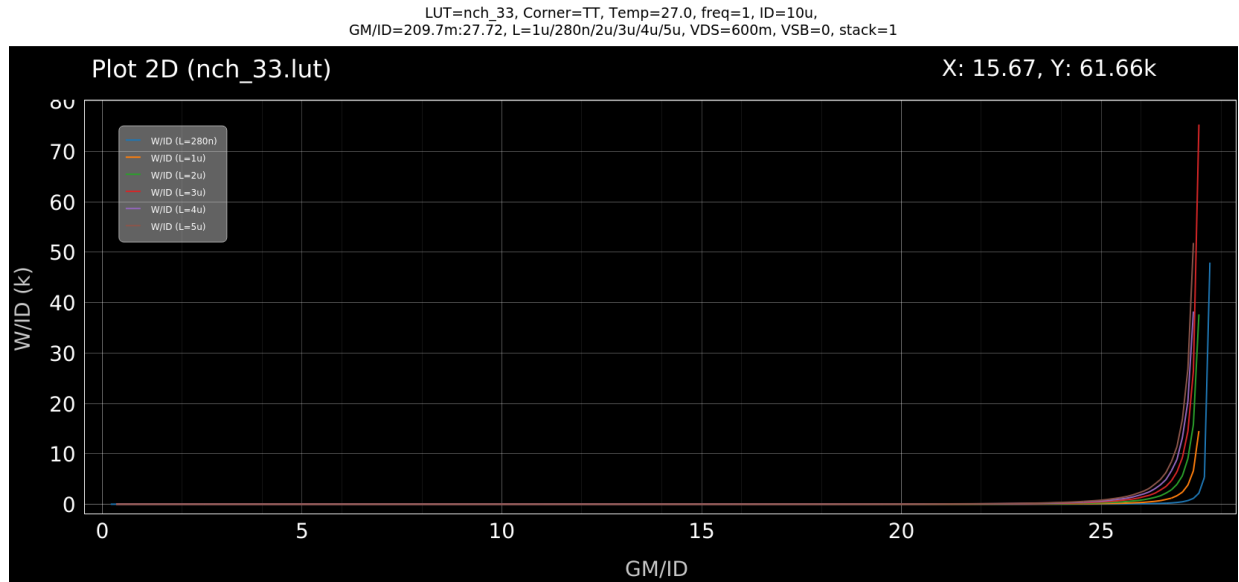


Figure 4: NMOS W/I_D vs. g_m/I_D .

1.2 PMOS

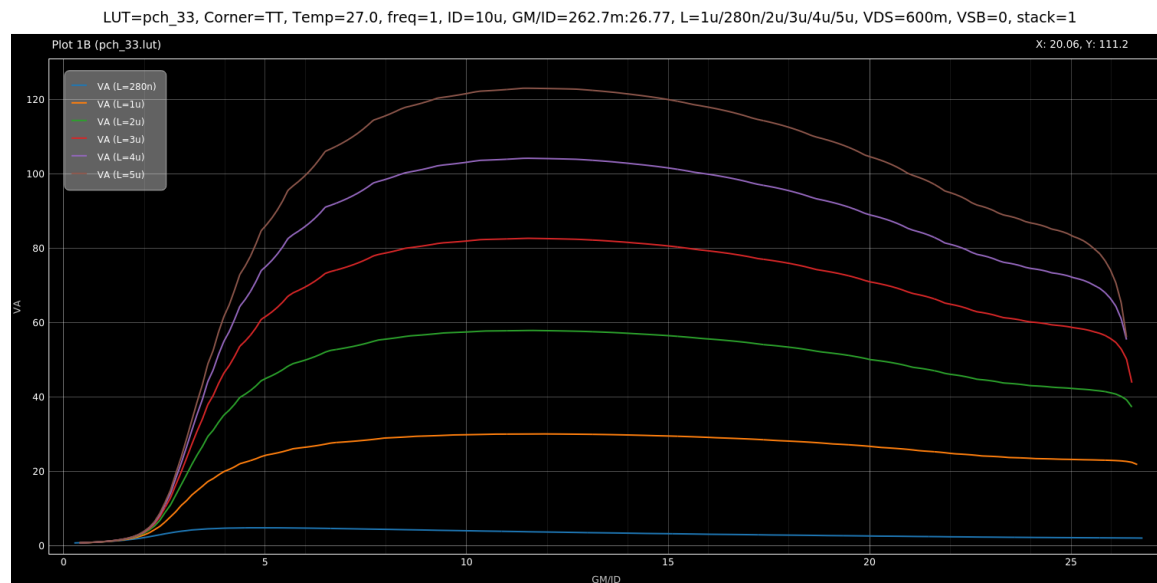


Figure 5: PMOS (V_A) vs. g_m/I_D .

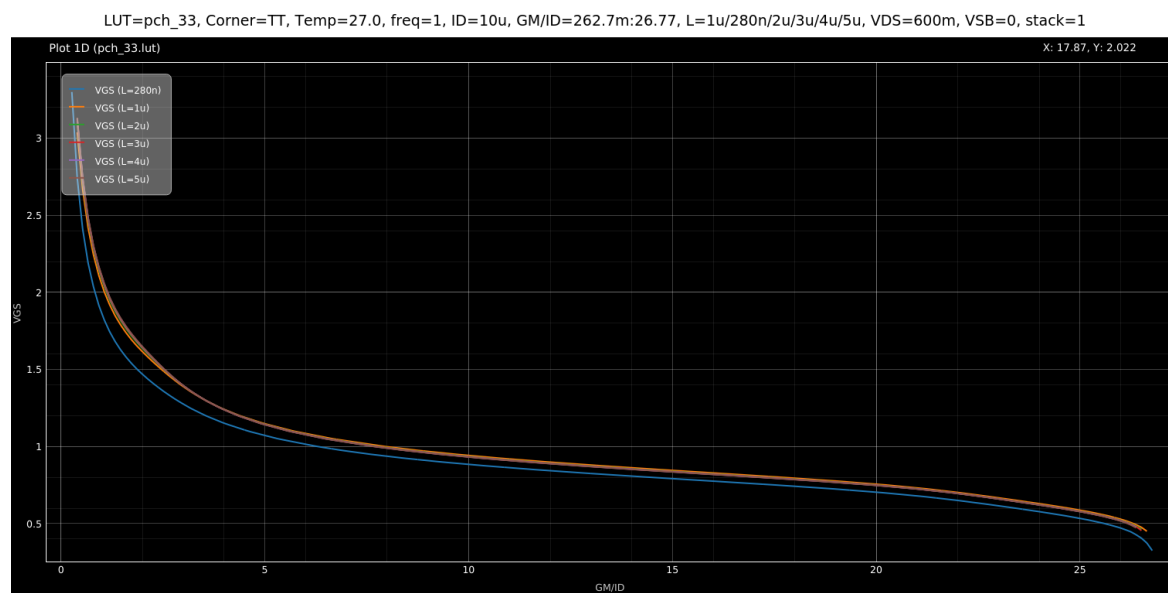


Figure 6: PMOS V_{GS} vs. g_m/I_D .

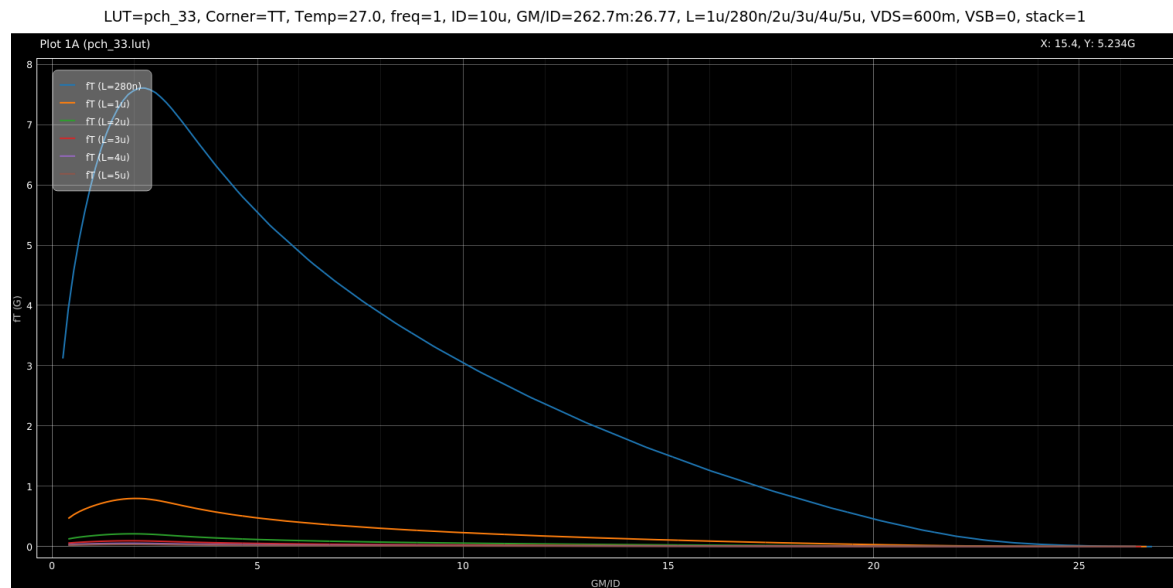


Figure 7: PMOS Transit Frequency (f_T) vs. g_m/I_D

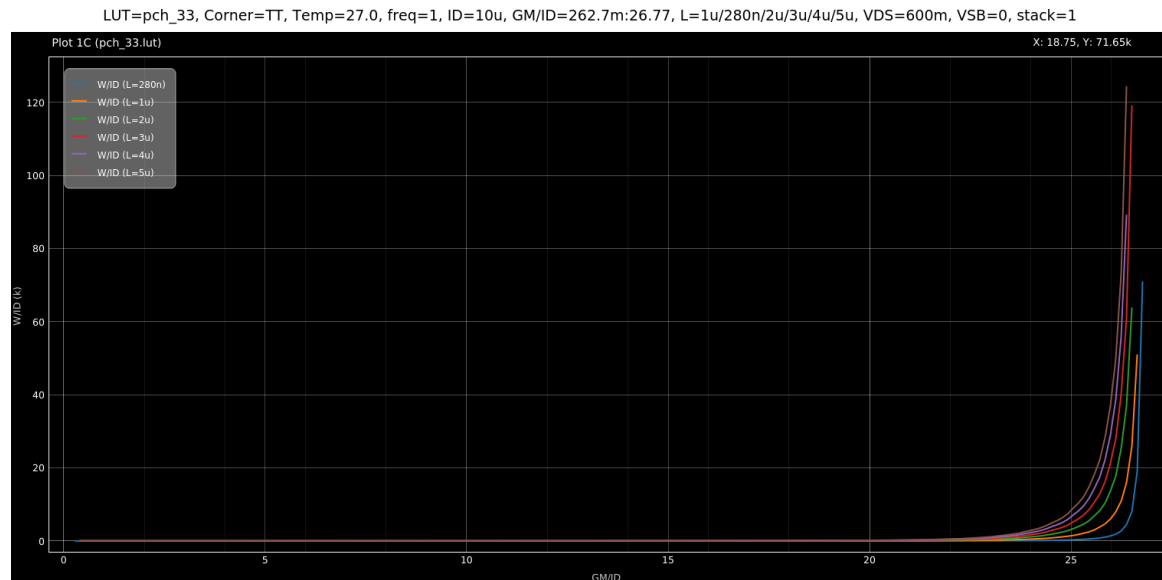


Figure 8: PMOS W/I_D vs. g_m/I_D .

2 PART 2: OTA Design

2.1 Specs:

Parameter	Specification (GF180MCU)
Supply Voltage (V_{DD})	2.5 V
Closed Loop Gain (A_{CL})	2 V/V
Phase Margin (at target A_{CL})	$\geq 70^\circ$
CM Input Range - Low	≤ 0 V
CM Input Range - High	≥ 1 V
Differential Output Swing	1.2 V peak-to-peak
Load Capacitance (C_L)	500 fF
DC Loop Gain	60 dB
Closed-Loop Settling Time (1% error)	100 ns

First, we will choose pmos input pair since the CM input low is 0V.

Taking the loading effect into consideration:

$$\beta = \frac{C_F}{C_F + C_{IN}} = \frac{1p}{1p + 2p} = \frac{1}{3}$$

$$LG = 60dB \rightarrow 1000 = A_{OL} * \beta \rightarrow A_{OL} = 3000$$

$$A_{OL} = \frac{C_{IN}}{C_F + C_{IN}} * A_o = \frac{2p}{1p + 2p} * A_o = \frac{2}{3} * A_o$$

$$BW_{OL} = BW_{OTA}$$

Gain-Bandwidth Product (GBW) Relationships

$$\left(\frac{C_{in}}{C_{in} + C_f} \right) A_o \cdot BW_{OTA} = A_{cl} \cdot BW_{cl}$$

$$GBW_{OTA} = 3 \cdot BW_{cl}$$

$$GBW_{OTA} = \frac{g_m}{2\pi C_{out}}$$

$$C_{out} = C_L + (C_f \parallel C_{in}) = 1.167 \text{ pF}$$

Using the 1% error settling time:

$$e^{-\frac{T}{\tau}} * 100 = 1 \rightarrow \tau = 21.7ns$$

$$BW_{CL} = \frac{1}{2\pi\tau} = 7.33MHz$$

$$GBW_{CL} = A_{CL} * BW_{CL} = \frac{BW_{CL}}{\beta} = 22MHz = \frac{gm_{in}}{2\pi(C_{out,tot})} \rightarrow gm_{in} \approx 161\mu S$$

We will take $gm_{in} \approx 165\mu S$

Output Resistance is $R_{out} = (r_{o3}gm_3(r_{o2} \parallel r_{o3})) \parallel (r_{o4}gm_4r_{o5})$

2.2 Sizing:

2.2.1 Input Pair:

Let $L = 300nm$ we will take min L to reduce C_{gg} .

$$gm_{in} \approx 165\mu S$$

Assume $gm/I_D = 16.5$

So we will get that the current is $I_D = 10\mu A$

$$W = 20.88 \mu m$$

LUT Settings		
ID	10u	?
gm/ID	16.5	?
L	0.3u	?
VDS	0.6	?
VSB	0	?
Stack	1	?
Results:		
Name	TT-27.0	
1 ID	10u	
2 IG	N/A	
3 L	300n	
4 W	20.88u	
5 VDB	600m	
6 VGB	782.4m	
7 VGS	782.4m	
8 VDS	600m	
9 VSB	0	
10 VGB	782.4m	

Figure 9: input pair

2.2.2 NMOS Current source:

choosing large L for bigger r_o to reduce noise and mismatch.

$$L = 1\mu m$$

And biasing in SI for lower gm and lower noise

$$\frac{gm}{I_D} = 8$$

we get

$$W = 5.52 \mu m$$

LUT Settings		
ID	20u	?
gm/ID	8	?
L	1u	?
VDS	0.6	?
VSB	0	?
Stack	1	?
Results:		
Name	TT-27.0	
1 ID	20u	
2 IG	N/A	
3 L	1u	
4 W	5.52u	
5 VDB	600m	
6 VGB	892.4m	
7 VGS	892.4m	
8 VDS	600m	
9 VSB	0	
10 VGB	892.4m	

Figure 10: NMOS CS

2.2.3 Cascode NMOS:

Choosing Large L= 1 μm and

$$\frac{gm}{ID} = 9$$

We get W = 3.56 μ

2.2.4 Cascode PMOS:

Choosing Large L= 1 μm and

$$\frac{gm}{ID} = 9$$

We get W = 15.3 μ

LUT Settings		
ID	10u	
gm/ID	9	
L	1u	
VDS	0.6	
VSB	0	
Stack	1	
Results:		
Name	TT-27.0	
1 ID	10u	
2 IG	N/A	
3 L	1u	
4 W	15.32u	
5 VDB	600m	
6 VGB	969.2m	
7 VGS	969.2m	
8 VDS	600m	
9 VSB	0	
10 VGB	969.2m	

Figure 12: Cascode Pmos

LUT Settings		
ID	10u	
gm/ID	9	
L	1u	
VDS	0.6	
VSB	0	
Stack	1	
Results:		
Name	TT-27.0	
1 ID	10u	
2 IG	N/A	
3 L	1u	
4 W	3.56u	
5 VDB	600m	
6 VGB	862.5m	
7 VGS	862.5m	
8 VDS	600m	
9 VSB	0	
10 VGB	862.5m	

Figure 11: Cascode Nmos

2.2.5 Tail PMOS:

Choosing Large L= 1 μm for lower mismatch and

$$\frac{gm}{ID} = 7$$

To lower gm for lower noise

We get W = 17.68 μ

2.2.6 PMOS Current Source:

Same as the tail

We get W = 8.84 μ

LUT Settings		
ID	10u	
gm/ID	7	
L	1u	
VDS	0.6	
VSB	0	
Stack	1	
Results:		
Name	TT-27.0	
1 ID	10u	
2 IG	N/A	
3 L	1u	
4 W	8.84u	
5 VDB	600m	
6 VGB	1.04	
7 VGS	1.04	
8 VDS	600m	
9 VSB	0	
10 VGB	1.04	

Figure 14: PMOS CS

LUT Settings		
ID	20u	
gm/ID	7	
L	1u	
VDS	0.6	
VSB	0	
Stack	1	
Results:		
Name	TT-27.0	
1 ID	20u	
2 IG	N/A	
3 L	1u	
4 W	17.68u	
5 VDB	600m	
6 VGB	1.04	
7 VGS	1.04	
8 VDS	600m	
9 VSB	0	
10 VGB	1.04	

Figure 13: Tail PMOS

Bias Voltages:

$$V_{CASCN} = V_{GScascN} + V_{MN}^* = 1.11V$$

$$V_{CASCN} = V_{DD} - V_{GScascP} - V_{MP}^* = 1.24V$$

Since we didn't take the body effect into consideration so we must Choose slighter higher and lower values to ensure transistor operate in saturation:

$$V_{CASCN} = 1.2V$$

$$V_{CASCN} = 1.15V$$

Table 1: Final Sizing

Transistor	Function	Type	ID (μA)	gm/ID	L (μm)	W (μm)
M1	Input Pair	PMOS	10	16.5	0.3	20.88
M2	Sink Mirror	NMOS	20	8.0	1.0	5.52
M3	Cascode	NMOS	10	9.0	1.0	3.56
M4	Cascode	PMOS	10	9.0	1.0	15.32
M5	Current Source	PMOS	10	7.0	1.0	8.84
M6	Tail Source	PMOS	20	7.0	1.0	17.68

3 PART 3: Open-Loop OTA (Behavioral CMFB)

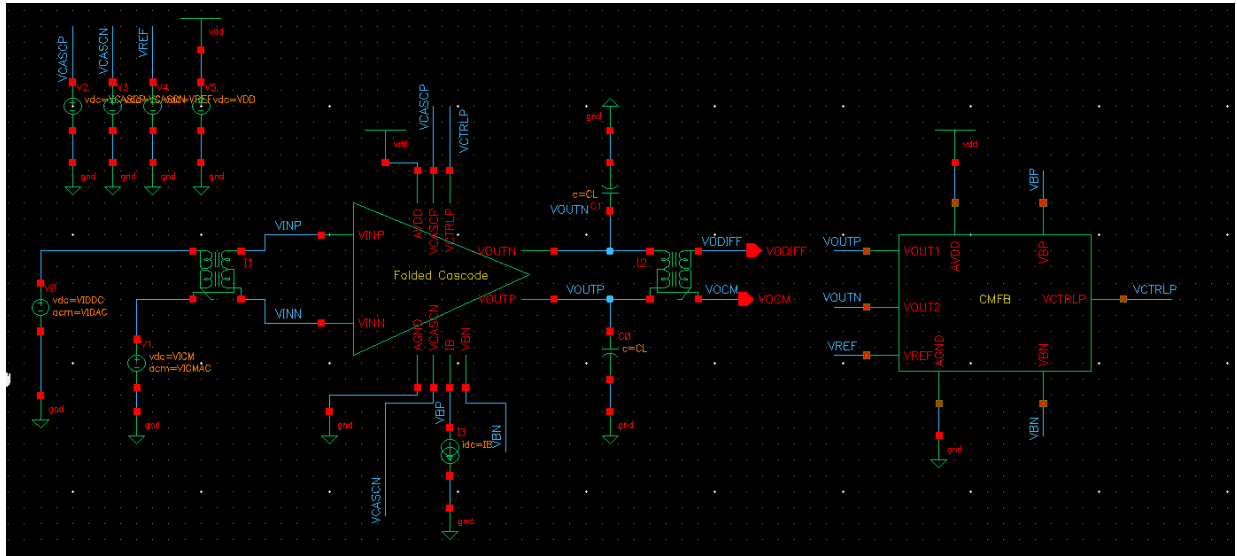


Figure 12: Open Loop Schematic

3.1 OP Analysis:

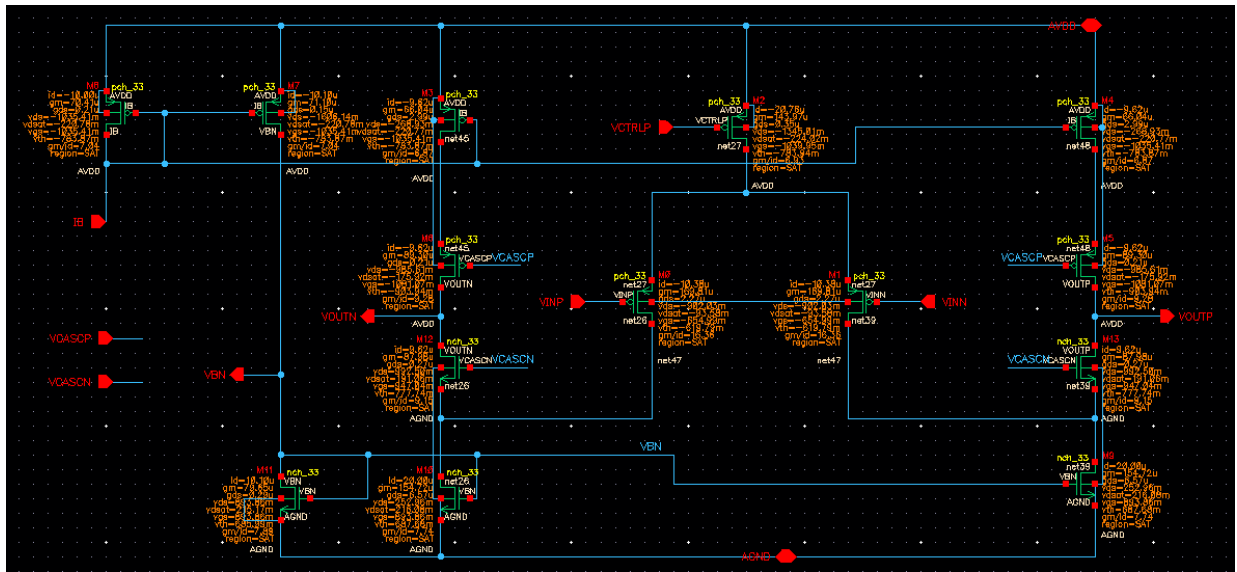


Figure 13: OP parameters before

Transistor(s)	Type	Function	ID (μA)	VGS	VDS	VDSAT	VTH	gm	gds	Region
M0, M1	PMOS	Input Pair	-10.42	-655.31	-981.58	-93.59	-620.78	170.27	2.29	SAT
M2	PMOS	Tail Source	-20.84	-1090.47	-1343.89	-224.40	-791.94	144.23	0.35	SAT
M3, M4	PMOS	Top Sources	-9.70	-1035.41	-303.95	-220.77	-753.87	67.40	1.06	SAT
M5, M6	PMOS	Cascodes	-9.70	-1046.05	-851.12	-161.47	-913.14	109.96	0.55	SAT

M9, M10	NMOS	Bottom Sinks	20.11	893.86	274.73	216.08	687.69	157.18	4.28	SAT
M12, M13	NMOS	Cascodes	9.70	925.27	970.28	169.18	728.73	100.92	4.72	SAT
M8	PMOS	Bias Diode	-10.00	-1035.41	-1035.41	-220.78	-753.87	70.41	0.21	SAT
M7	PMOS	Bias Mirror	-10.10	-1035.41	-1006.14	-220.78	-753.87	71.10	0.15	SAT
M11	NMOS	Bias Diode	10.10	893.86	893.86	215.17	685.09	79.65	0.29	SAT

ITI_LABS:Folded_tb:1	v("/VOCM" ?result "dcOp")	1.245
ITI_LABS:Folded_tb:1	v("/VOUTN" ?result "dcOp")	1.245
ITI_LABS:Folded_tb:1	v("/VOUTP" ?result "dcOp")	1.245
ITI_LABS:Folded_tb:1	v("/CTRLP" ?result "dcOp")	1.46
ITI_LABS:Folded_tb:1	v("/VBP" ?result "dcOp")	1.465

What is the CM level at the OTA output?

(V_{OCM}) is 1.245V, which is the reference voltage (V_{REF}) to approximately half of the 2.5V supply.

What are the differential input and output voltages of the error amplifier? What is the relation between them?

The inputs of CMFB are common-mode voltage (V_{CM}) and the target reference (V_{REF}). $V_{diff} = 5mV$. Its output is V_{CTRLP} . Because the ideal error amplifier is assigned a gain of 1, the relationship is defined by:

$$V_{CTRLP} = V_{BP} + 1 \cdot (V_{CM} - V_{REF})$$

The simulation confirms this: with $V_{OCM} = 1.245V$, the generated V_{CTRLP} is at 1.46V, which is almost perfectly aligned with the bias voltage V_{BP} of 1.465V.

3.2 Diff small signal ccs:

ITI_LABS:Folded_tb:1	mag(VF("/VODIFF"))	
ITI_LABS:Folded_tb:1	VODIFF_dB	
ITI_LABS:Folded_tb:1	Ao	6.595k
ITI_LABS:Folded_tb:1	Ao_dB	76.38
ITI_LABS:Folded_tb:1	BW	7.424k
ITI_LABS:Folded_tb:1	UGF	49.07M
ITI_LABS:Folded_tb:1	GBW	48.96M
ITI_LABS:Folded_tb:1	phaseMargin(VF("/VODIFF"))	-95.81

Figure 14: Ac parameters before resizing

As we see the open loop gain is very high more than required (3000), to reduce it we need to reduce Rout.

For the cascode devices we will reduce gm and ro so we will reduce the W and L

For Cascode NMOS:

Reduce W from 3.56u to 2.54u and L from 1u to 0.5u.

For Cascode PMOS:

Reduce W from 15.3u to 10u and L from 1u to 0.5u.

VREF	1.25					
VICM	0.5					
VICMAC	0					
VIDAC	1					
VIDDC	0					
Ln	0.5u					
Lp	0.5u					
Wn	2.5u					
Wp	10u					
sdd variable						

Test	Output	Nominal	Spec	Weight	Pass/Fail
ITI_LABS:Folded_tb:1	mag(VF("VODIFF"))				
ITI_LABS:Folded_tb:1	VODIFF_dB				
ITI_LABS:Folded_tb:1	Ao	4.063k			
ITI_LABS:Folded_tb:1	Ao_dB	72.18			
ITI_LABS:Folded_tb:1	BW	12.46k			
ITI_LABS:Folded_tb:1	UGF	50.77M			
ITI_LABS:Folded_tb:1	GBW	50.62M			
ITI_LABS:Folded_tb:1	phaseMargin(VF("VODIFF"))	-94.43			

Figure 15: AC parameters after resizing

The PM is $= 180 - 94 = 85$

Also I checked the OP parameters again

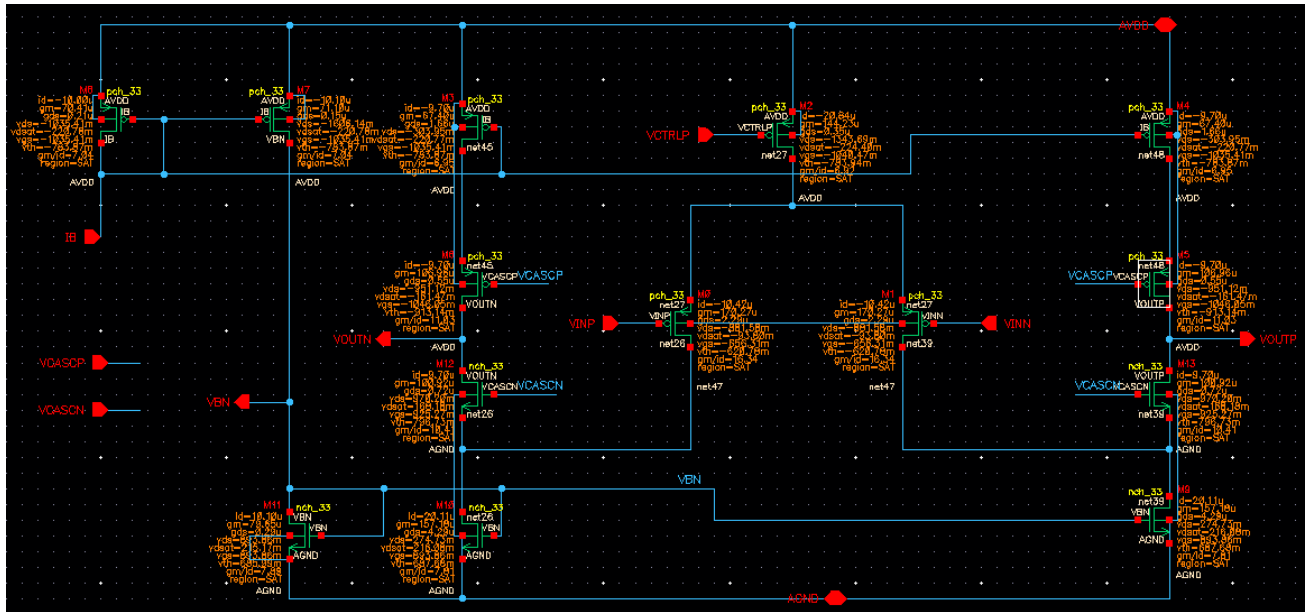


Figure 16: OP after resizing

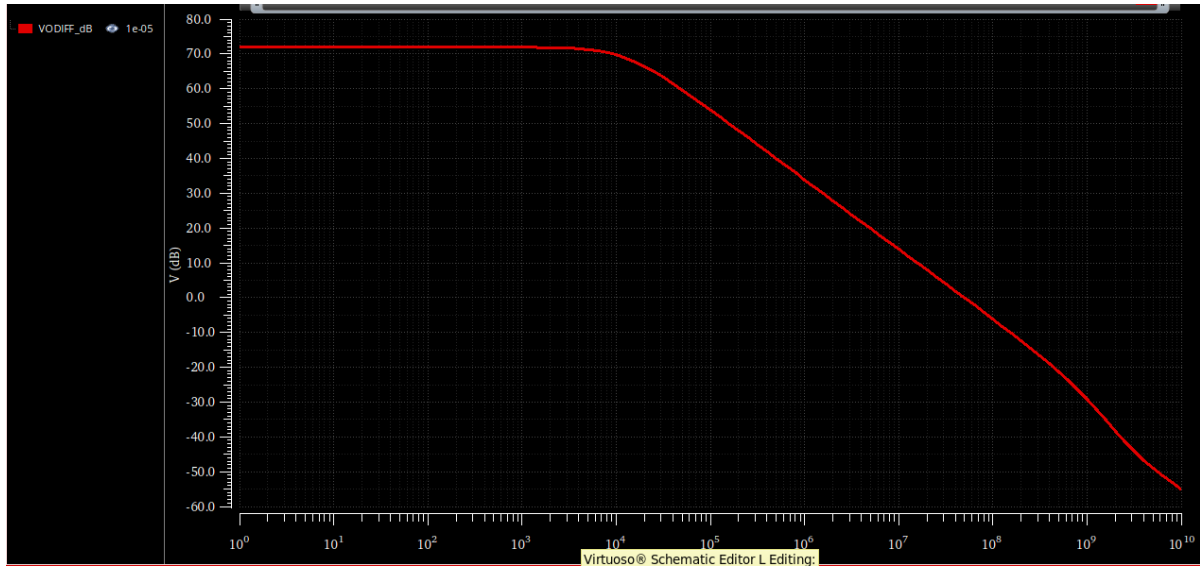


Figure 17: Differential open-loop magnitude response (dB) vs. frequency.

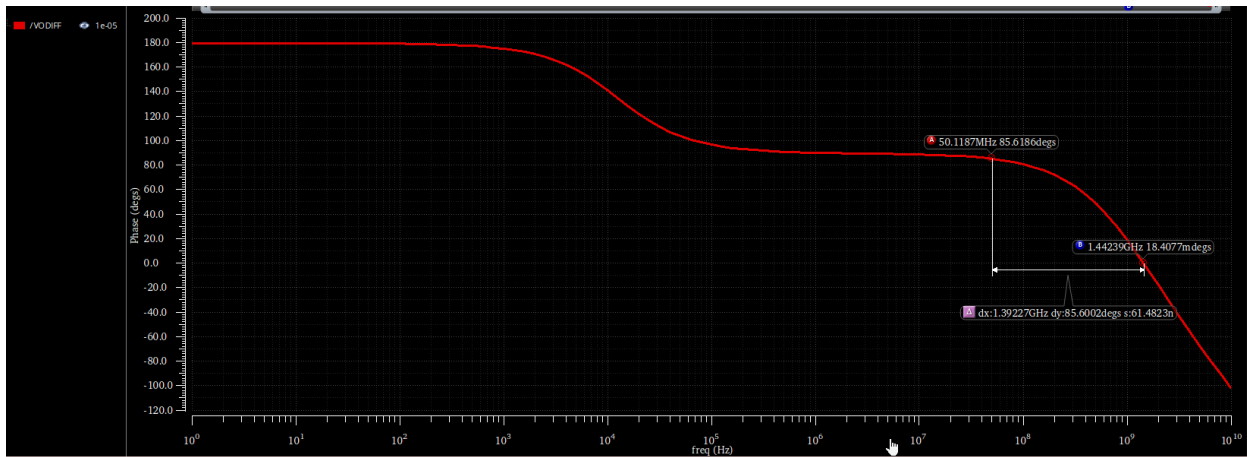


Figure 18: Differential open-loop phase response vs. frequency.

$$PM = 85^\circ$$

Hand Analysis:

$$R_{OUT} = [gm_{casen}ro_{casen}(r_{mn}/ro_{inp})]//gm_{cascp}ro_{cascp}ro_{mp} \approx 19M\Omega$$

$$A_{OL} = GmR_{OUT} = 170\mu * 19M = 3.23K$$

$$BW_{OL} = \frac{1}{2\pi R_{OUT}C_{OUT}} \approx 16.75KHz$$

$$GBW_{OL} = UGF = A_{OL} * BW_{OL} = 54MHz$$

	Simulation	Hand Analysis
Aol	72.18 dB (4063 V/V)	3230
BW	12.46 kHz	16.75KHz
GBW	50.62 MHz	454MHz
UGF	50.77 MHz	54MHz

4 PART 4: Open-Loop OTA Simulation (Actual CMFB)

4.1 CMFB Design:

I designed the CMFB circuit to have 25% of the current budget of the OTA which is 10uA so every branch will have current = 2.5 uA.

First lets estimate the sensing resistance , in worst case the $\Delta V_{out} = 1.2$ if I choose the $R = 50k$ as in the lab which makes $I_x = 1.2/100k = 12u$ we must have a current in the common drain larger than 12 u but I designed to make it 2.5u so we will increase the R to be 250k instead of 50k that makes $I_x = 2.4uA$ and that's fine.

$$R \geq \frac{0.6}{2.5u}$$

$$R \geq 240k$$

I will take $R = 250k$

The Pmos current sources will have a width $W = 2.21u$ and $L = 1u$ which is half width of the pmos CS of main OTA.

For the tail NMOS CS will have a width equals the quarter of the NMOS CS of the OTA $W=1.38u$

For the NMOS inputs of the differential pair we will sweep over W to make the gm of the diff pair equals the gm of the active loads to have a gain of 1.

We found the equal gm at $W = 0.6u$.

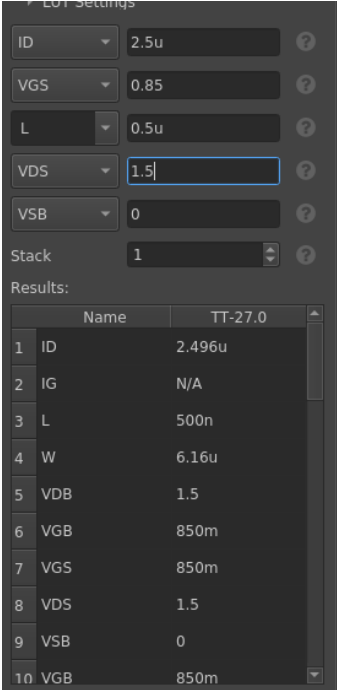
Also since V_{out} had a positive shift because of the Common drain , so we will shift the V_{ref} by the same shift so we will use same sizing.

Calculating new Output Swing:

$$\text{minimum Output Swing} = V_{CASCN}^* + V_{MN}^* \approx 0.5$$

$$\text{maximum Output Swing} = V_{DD} - V_{GSP} - V_p^* = 1.5$$

$$\text{Middle of Output Swing} = V_{REF_{new}} \approx 1V$$



LOT Settings		
ID	2.5u	?
VGS	0.85	?
L	0.5u	?
VDS	1.5	?
VSB	0	?
Stack	1	?
Results:		
Name	TT-27.0	
1 ID	2.496u	
2 IG	N/A	
3 L	500n	
4 W	6.16u	
5 VDB	1.5	
6 VGB	850m	
7 VGS	850m	
8 VDS	1.5	
9 VSB	0	
10 VGB	850m	

4.2 OP Analysis:

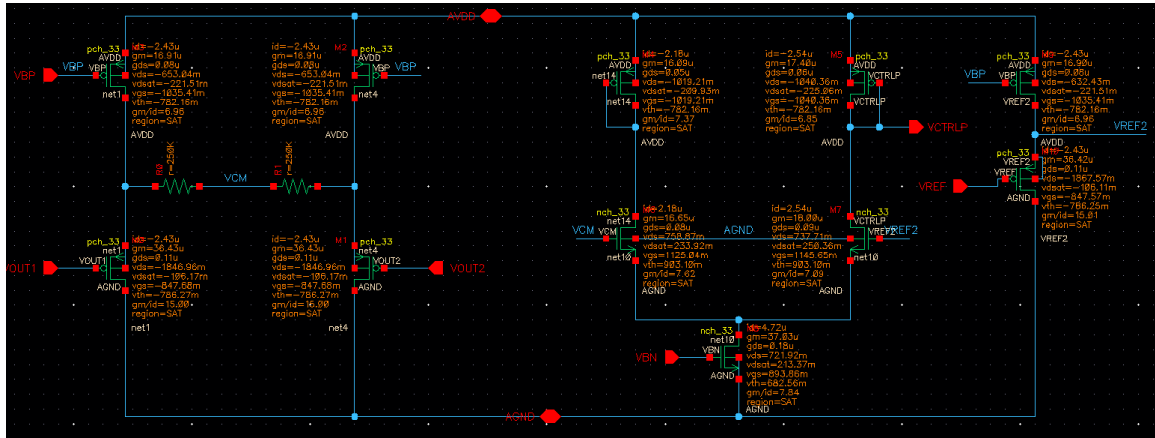


Figure 19: DC operating point of the actual CMFB

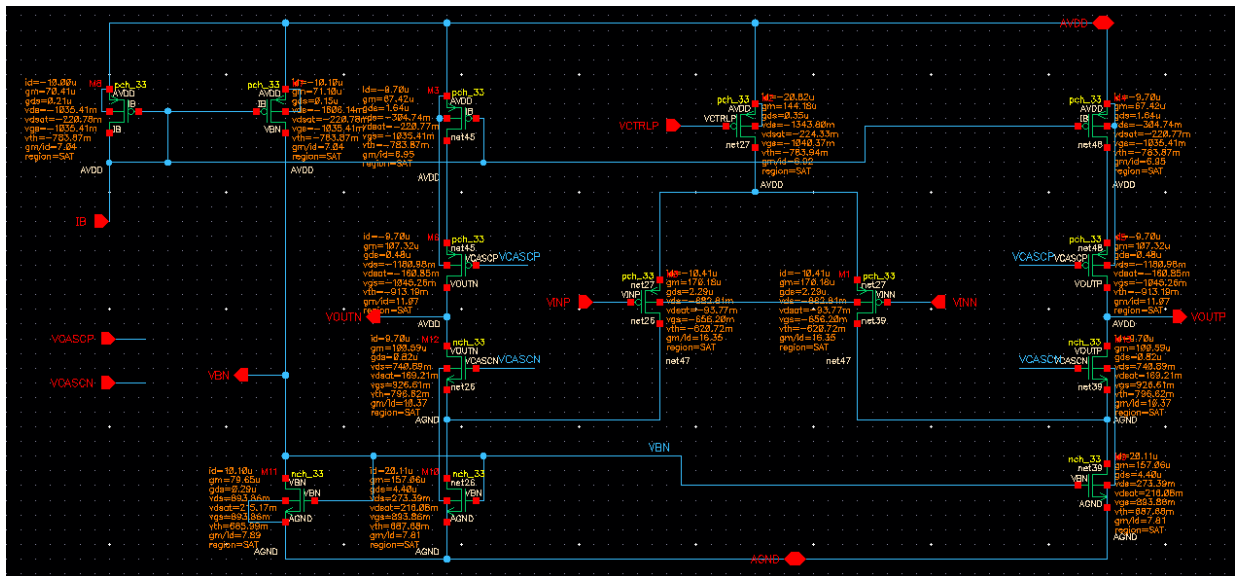


Figure 20: DC operating point of the main folded-cascode OTA

ITI_LABS:Folded_tb:1	v("VOCM" ?result "dcOp")	1.005
ITI_LABS:Folded_tb:1	v("VOUTN" ?result "dcOp")	1.005
ITI_LABS:Folded_tb:1	v("VOUTP" ?result "dcOp")	1.005
ITI_LABS:Folded_tb:1	v("VCTRLP" ?result "dcOp")	1.46
ITI_LABS:Folded_tb:1	v("VBP" ?result "dcOp")	1.465
ITI_LABS:Folded_tb:1	v("/I02/VREF2" ?result "dcOp")	1.868
ITI_LABS:Folded_tb:1	v("/I02/VCM" ?result "dcOp")	1.853

Figure 21: VOCM where V_{REF2} is V_{REF} after shifting

Table 2: Main OTA & Biasing DC Operating Point

Transistor(s)	Function	ID (μ A)	VGS (mV)	VDS (mV)	VDSAT (mV)	VTH (mV)	gm (μ S)	gds (μ S)	Region
M0, M1	Input Pair	10.41	656.20	882.81	93.77	620.72	170.18	2.29	SAT
M2	Tail Source	20.82	1040.37	1343.80	224.33	783.94	144.18	0.35	SAT
M3, M4	Top Sources	9.70	1035.41	304.74	220.77	753.87	67.42	1.64	SAT
M5, M6	Cascodes	9.70	1045.26	1180.98	160.85	913.18	107.32	0.48	SAT
M9, M10	Bottom Sinks	20.11	893.86	273.39	216.06	687.68	157.06	4.40	SAT
M12, M13	Cascodes	9.70	926.61	740.89	169.21	798.62	100.59	0.82	SAT
M8	Bias Diode	10.00	1035.41	1035.41	-220.78	753.87	70.41	0.21	SAT
M7	Bias Mirror	10.10	1035.41	1006.14	220.78	753.87	71.10	0.15	SAT
M11	Bias Diode	10.10	893.86	893.86	215.17	685.09	79.65	0.29	SAT

Transistor(s)	Function	ID (μ A)	VGS (mV)	VDS (mV)	VDSAT (mV)	VTH (mV)	gm (μ S)	gds (μ S)	Region
CD Buffers	Level Shifters	2.43	847.68	1846.96	106.17	786.27	36.43	0.11	SAT
CD Bias	Buffer Sources	2.43	1035.41	653.04	221.51	782.16	16.91	0.08	SAT
Err Amp In	Diff Pair	2.18	1125.04	758.87	233.92	903.10	16.65	0.08	SAT
Err Amp Load	Diode Loads	2.18	1019.21	1019.21	209.93	782.16	16.09	0.05	SAT
Err Amp Tail	Tail Source	4.72	893.86	721.92	213.37	682.56	37.03	0.18	SAT
VREF Buffer	Level Shifter	2.43	847.57	1867.57	106.11	786.25	36.42	0.11	SAT

Table 3: CMFB Circuit DC Operating Point

What is the CM level at the OTA output? Why?

$V_{oCM} = 1.005$ which is almost equal to $V_{ref}=1$, the CMFB senses the average and compares it to V_{ref} and return to the tail CS forcing V_{ocm} to follow V_{ref} .

What are the differential input and output voltages of the error amplifier? What is the relation between them?

$$VDIFF_{IN} = 1.868 - 1.853 = 15mV$$

$$VDIFF_{OUT} = 1.465 - 1.46 = 5mV$$

$$\frac{VDIFF_{OUT}}{VDIFF_{IN}} = 0.333$$

4.3 Diff small signal ccs:

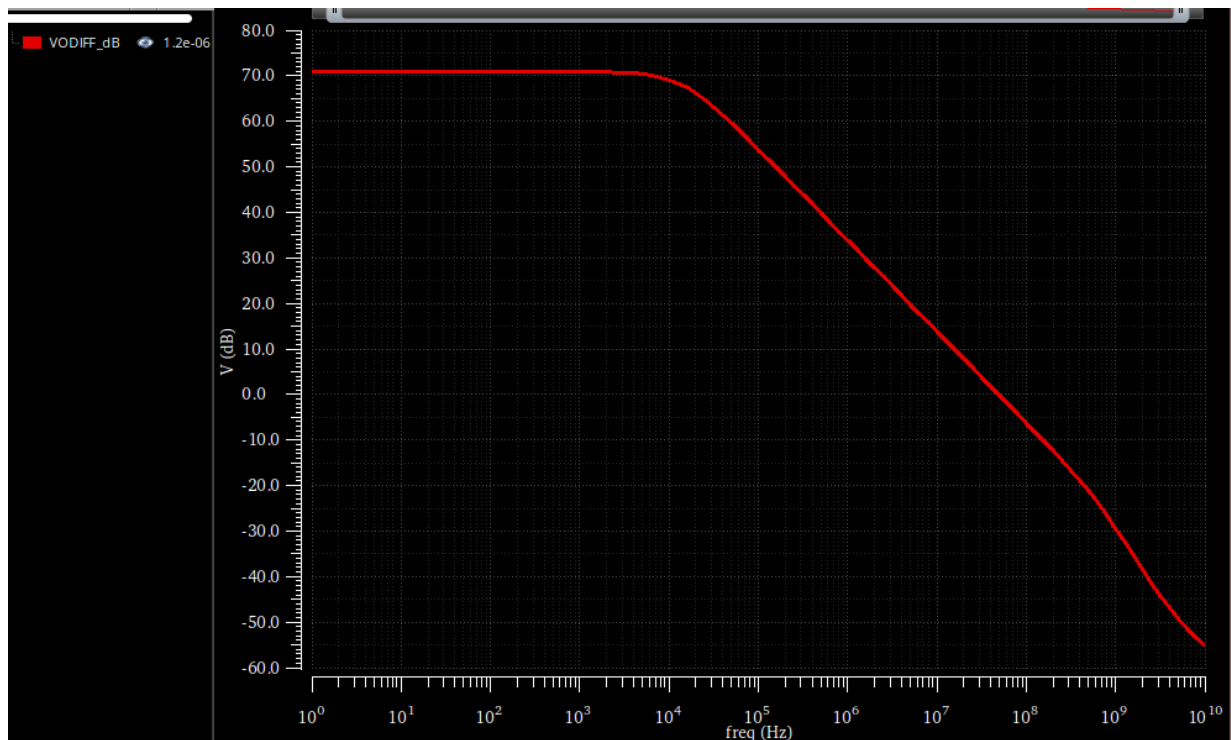


Figure 22: Differential open-loop magnitude response (dB) with actual CMFB loading

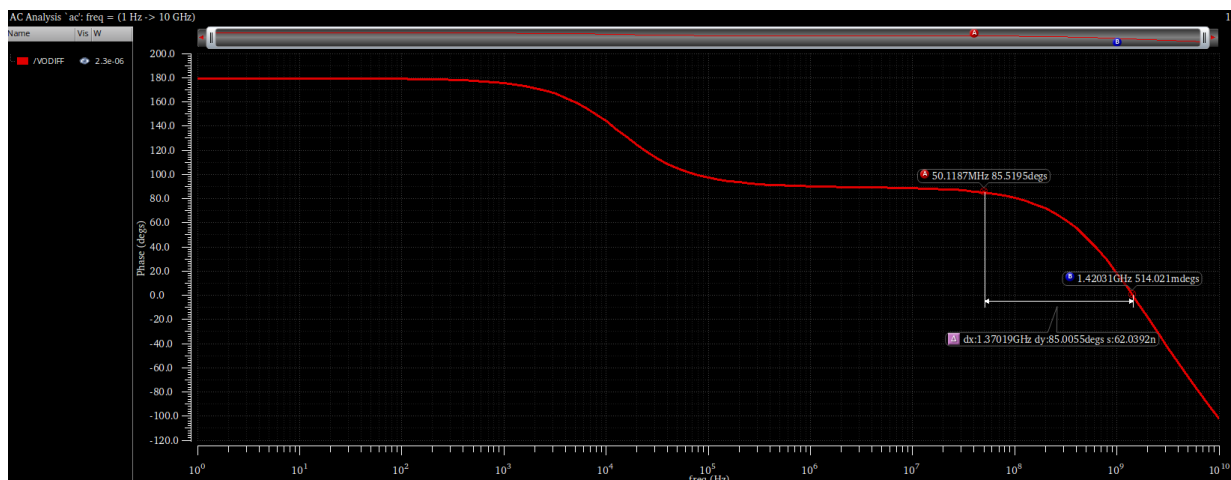


Figure 23: Differential open-loop phase response with actual CMFB loading

ITI_LABS:Folded_tb:1	mag(VF("/VODIFF"))	
ITI_LABS:Folded_tb:1	VODIFF_dB	
ITI_LABS:Folded_tb:1	Ao	3.615k
ITI_LABS:Folded_tb:1	Ao_dB	71.16
ITI_LABS:Folded_tb:1	BW	13.92k
ITI_LABS:Folded_tb:1	UGF	50.2M
ITI_LABS:Folded_tb:1	GBW	50.31M
ITI_LABS:Folded_tb:1	phaseMargin(VF("/VODIFF"))	-94.42

Figure 24: AC Values

$PM = 180 - 94 = 85^\circ$

Parameter	Behavioral CMFB	Actual CMFB
DC Gain (A_0)	72.18 dB	71.16 dB (3.615k V/V)
-3dB Bandwidth (BW)	12.46 kHz	13.92 kHz
Unity Gain Freq. (UGF)	50.77 MHz	50.20 MHz
Gain-Bandwidth (GBW)	50.62 MHz	50.31 MHz
Phase Margin (PM)	85.6°	85.5°

5 PART 5: Closed Loop Simulation (AC and STB)

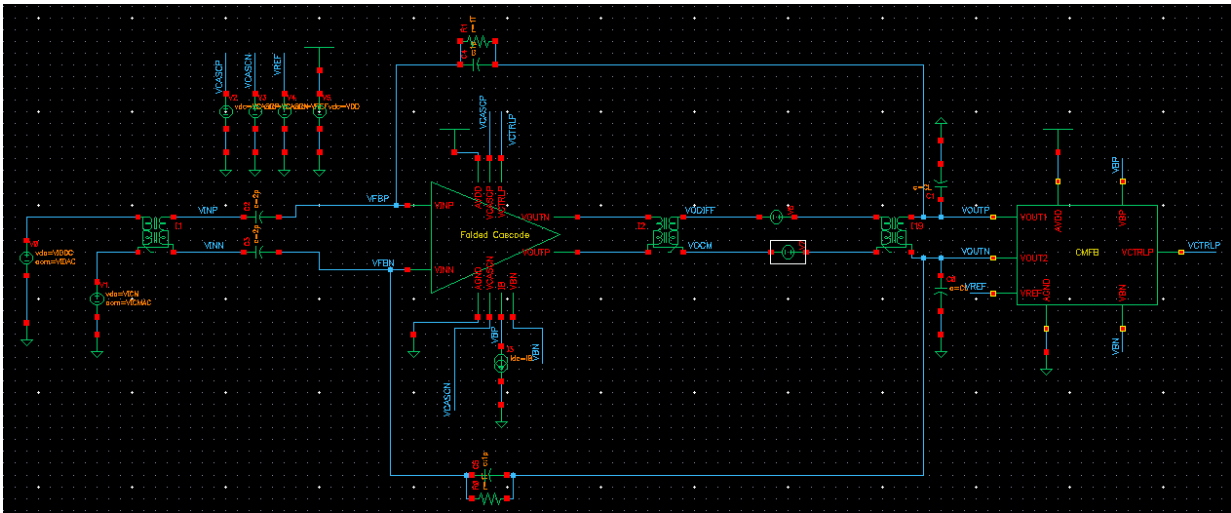


Figure 25: Closed Loop Schematic

5.1 OP Analysis:

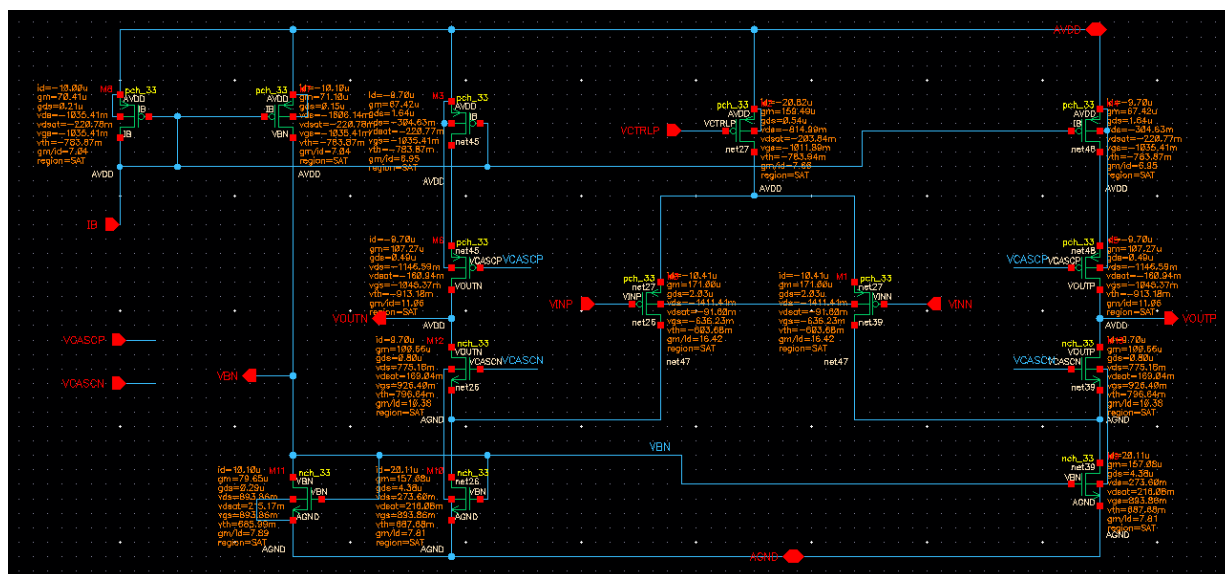


Figure 26: Folded OP

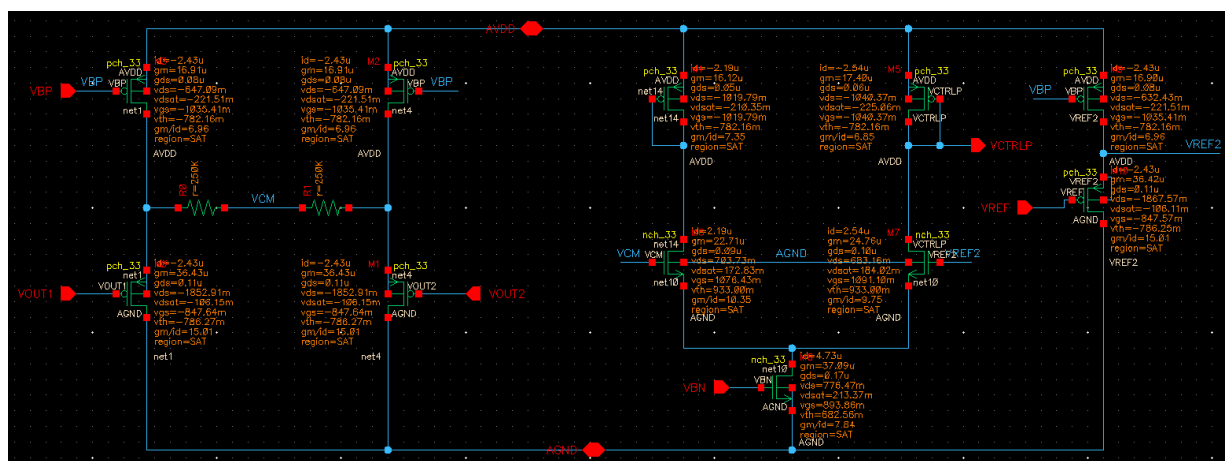


Figure 27: CMFB OP

ITI_LABS:Folded_tb_cl:1	v("/VOCM" ?result "dcOp")	1.003
ITI_LABS:Folded_tb_cl:1	v("/VFBP" ?result "dcOp")	1.003
ITI_LABS:Folded_tb_cl:1	v("/VFBN" ?result "dcOp")	1.003

Figure 28: OP

What is the CM level at the OTA output? Why?

It equals to 1.003 almost 1 as Vref.

What is the CM level at the OTA input? Why?

Also It equals to 1.003 almost 1 as V_{ref} . Follows the output by the feedback.

5.2 Differential closed-loop response:

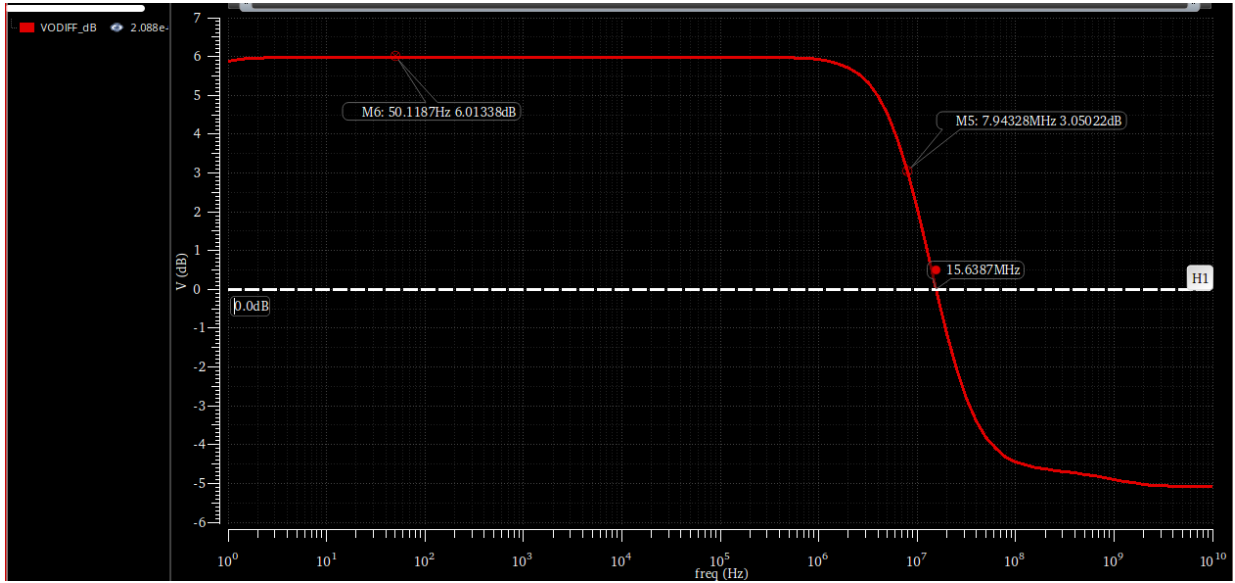


Figure 29: Closed-loop differential magnitude response (dB).

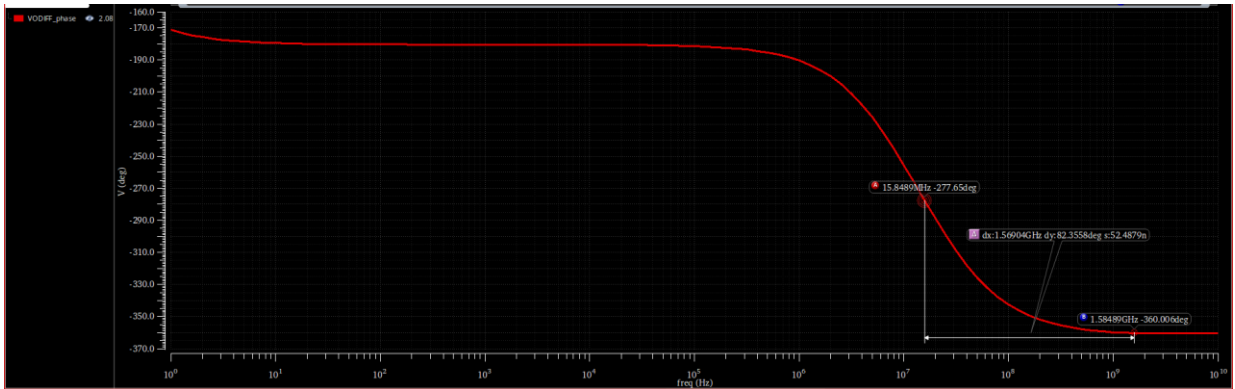


Figure 30: Closed-loop differential phase response

IT1_LABS:Folded_tb_d:1	VF("VODIFF")	
IT1_LABS:Folded_tb_d:1	mag(VF("VODIFF"))	
IT1_LABS:Folded_tb_d:1	VODIFF_dB	
IT1_LABS:Folded_tb_d:1	VODIFF_phase	
IT1_LABS:Folded_tb_d:1	Ao	1.998
IT1_LABS:Folded_tb_d:1	Ao_dB	6.013
IT1_LABS:Folded_tb_d:1	BW	8.267M
IT1_LABS:Folded_tb_d:1	GBW	16.36M
IT1_LABS:Folded_tb_d:1	UGF	15.65M

Figure 31:AC

$$GBW_{CL} = GBW_{Loaded} * \frac{2}{3}$$

$$GBW_{loaded} = 23 \text{ MHz} = \frac{g_m}{2\pi C_{tot}} \quad \text{from part 4}$$

SO

$$GBW_{CL} = 15.33 \text{ MHz}$$

So our GBW meets the specs

$$\left(\frac{C_{in}}{C_{in} + C_f} \right) A_o \cdot BW_{OTA} = A_{cl} \cdot BW_{cl}$$

$$BW_{cl} = \frac{GBW_{loaded}}{3} = 7.67 \text{ MHz}$$

5.3 Differential and CMFB loops stability (STB analysis):

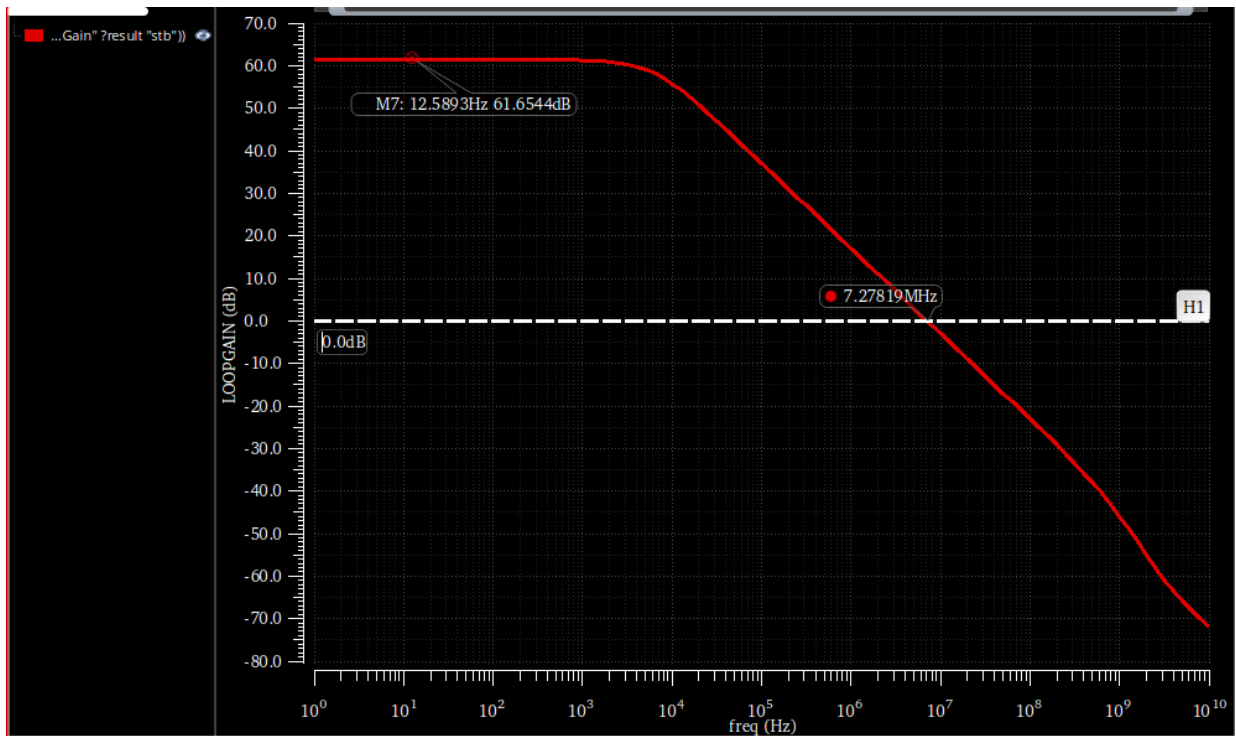


Figure 32: Differential Loop Gain magnitude (dB).

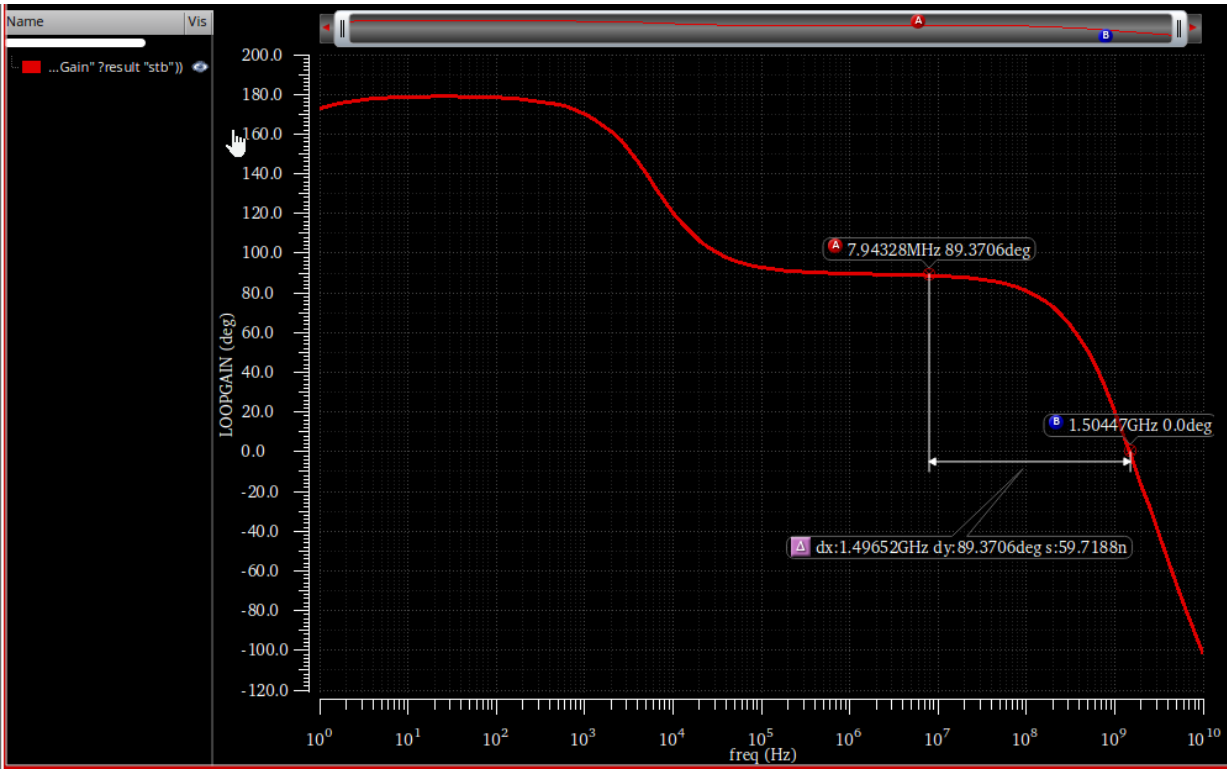


Figure 33: Differential Loop Gain phase

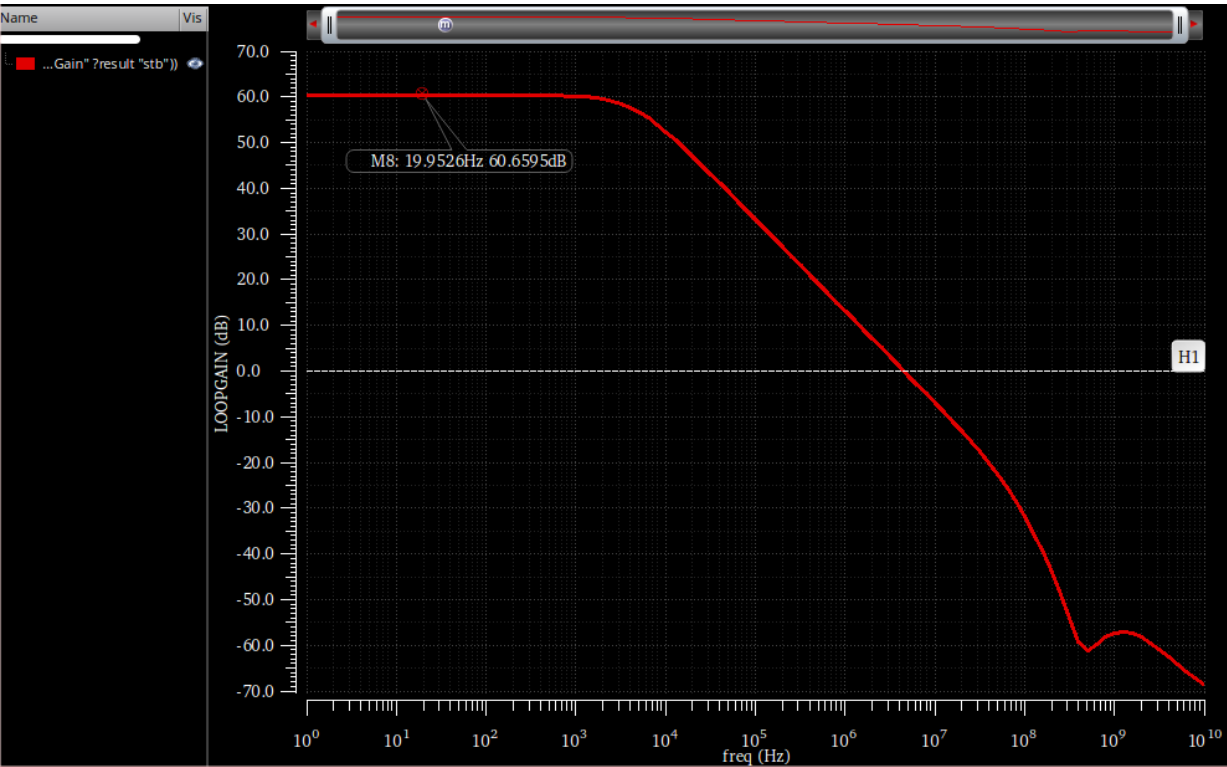


Figure 34: Common-Mode Loop Gain magnitude (dB)

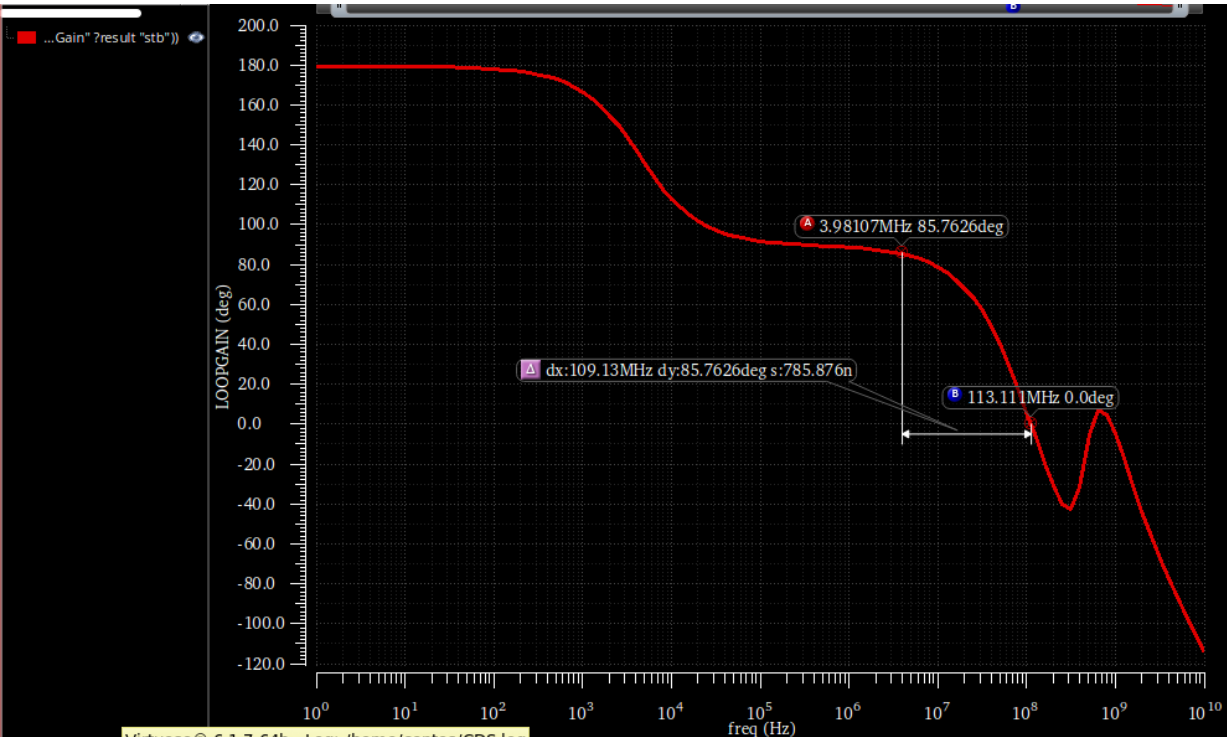


Figure 35:Common-Mode Loop Gain phase

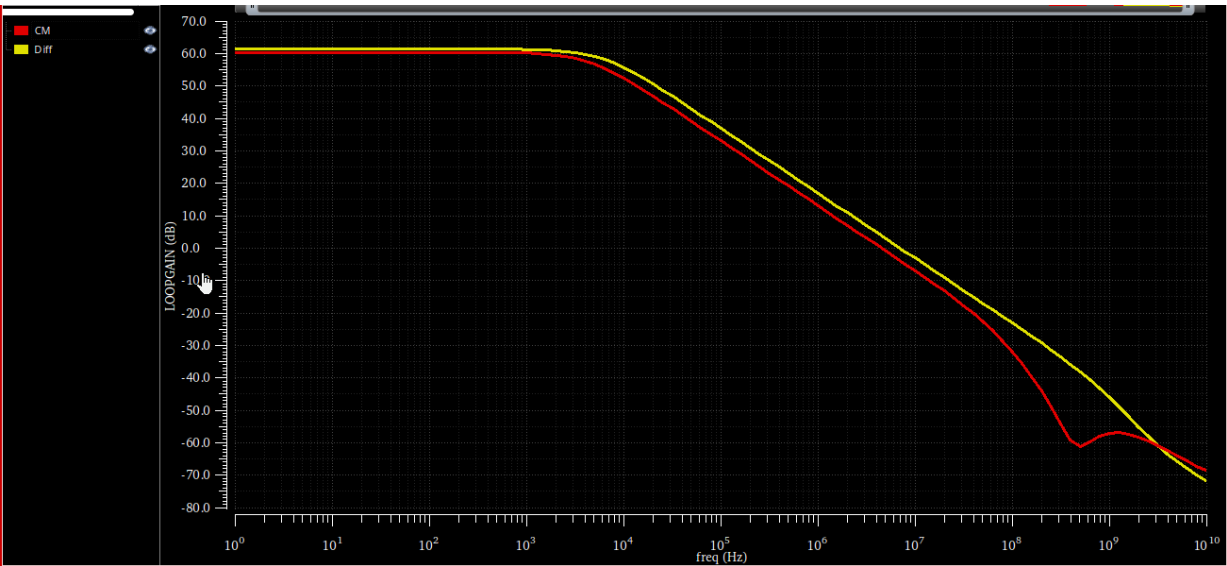


Figure 36: LG overlaid

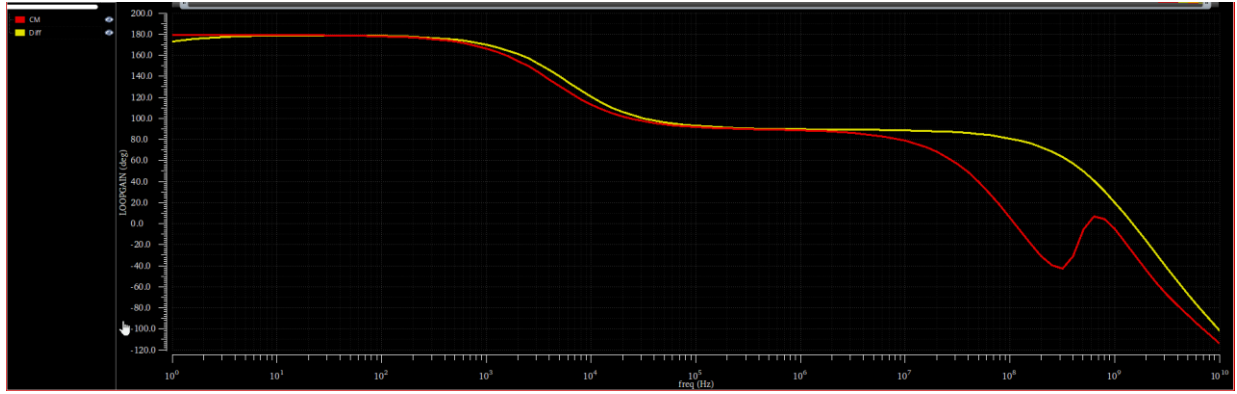


Figure 37: LG phase overlaid

ITI_LABS:Folded_tb_ci:1	db20(getData("loopGain" ?resul..	
ITI_LABS:Folded_tb_ci:1	phase(getData("loopGain" ?resu...	
ITI_LABS:Folded_tb_ci:1	PM_LG	89.44
ITI_LABS:Folded_tb_ci:1	GBW_LG	7.168M
ITI_LABS:Folded_tb_ci:1	ymax(mag(getData("loopGain" ?...	1.223k

Figure 38: Differential Loop Gain

ITI_LABS:Folded_tb_ci:1	db20(getData("loopGain" ?resul..	
ITI_LABS:Folded_tb_ci:1	phase(getData("loopGain" ?resu...	
ITI_LABS:Folded_tb_ci:1	PM_LG	85.06
ITI_LABS:Folded_tb_ci:1	GBW_LG	4.63M
ITI_LABS:Folded_tb_ci:1	ymax(mag(getData("loopGain" ?...	1.079k

Figure 39: CM Loop Gain

Parameter	Differential Loop	Common-Mode Loop
DC Loop Gain (LG)	61.65 dB (1.223k V/V)	60.65 dB (1.079k V/V)
Loop Gain GBW	7.168 MHz	4.63 MHz
Phase Margin (PM)	89.44°	85.06°

Compare GBW and PM of diff and CM loops. Comment.

	Diff	CM
GBW	7.168	4.63
PM	89	85

The differential loop has a higher Loop Gain GBW (7.168 MHz) compared to the CM loop (4.63 MHz). This is expected. The differential signal path is short and driven by the massive g_m of the main input pair. The CM signal path travels through the CD and the error amplifier.

Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation.
Comment.

	LG	OL
GBW	7.168	50
DC LG	61.65 dB (1.223k V/V)	71dB

In the open-loop simulation (Part 4), the DC gain was 71.16 dB and the GBW was 50.31 MHz. In this STB simulation, the DC Loop Gain drops to 61.65 dB. This exact ≈ 9.5 dB drop perfectly matches the feedback factor (β) introduced by the capacitive divider: $\beta = C_f / (C_s + C_f) = 1\text{pF}/3\text{pF} = 1/3$, and $20 \log_{10}(1/3) = -9.54$ dB. The STB GBW (7.168 MHz) is the BWcl which is the $GBW_{loaded} \frac{(\approx 23)}{\beta_{Beta}} = 7.67\text{MHz}$.

6 PART 6: Closed Loop Simulation (Transient):

6.1 Differential and CMFB loops stability:

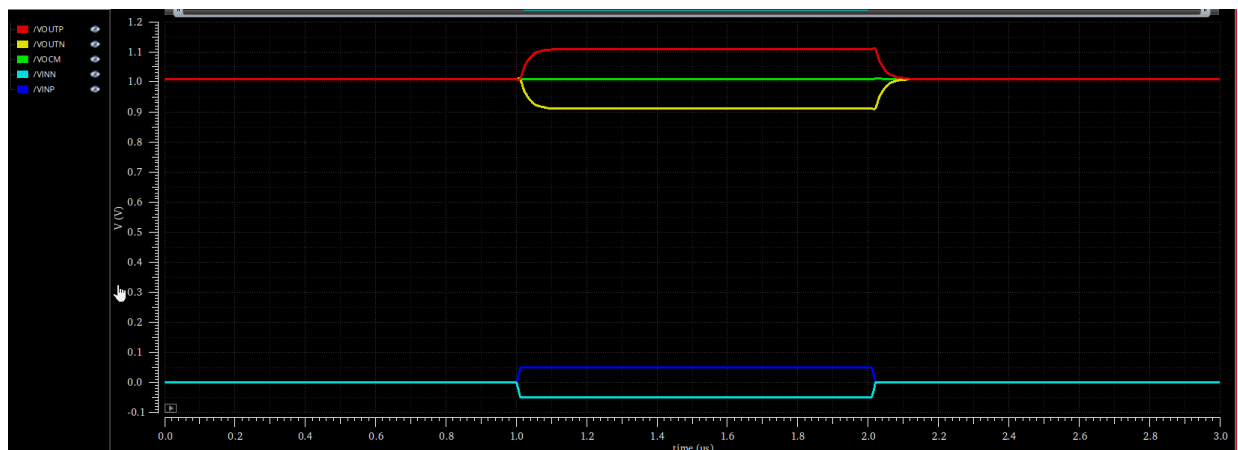
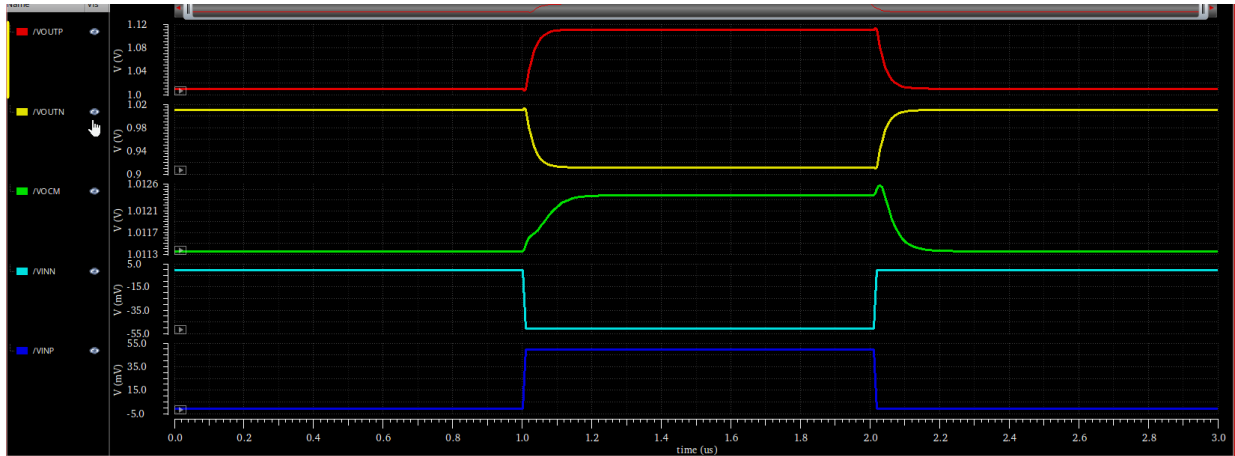


Figure 40: VINP, VINN, VOUTP, VOUTN, and VOCM overlaid

Figure 41: V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , and V_{OCM}

Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

No ringing just a little overshoot in the VOCM as we have high PM.

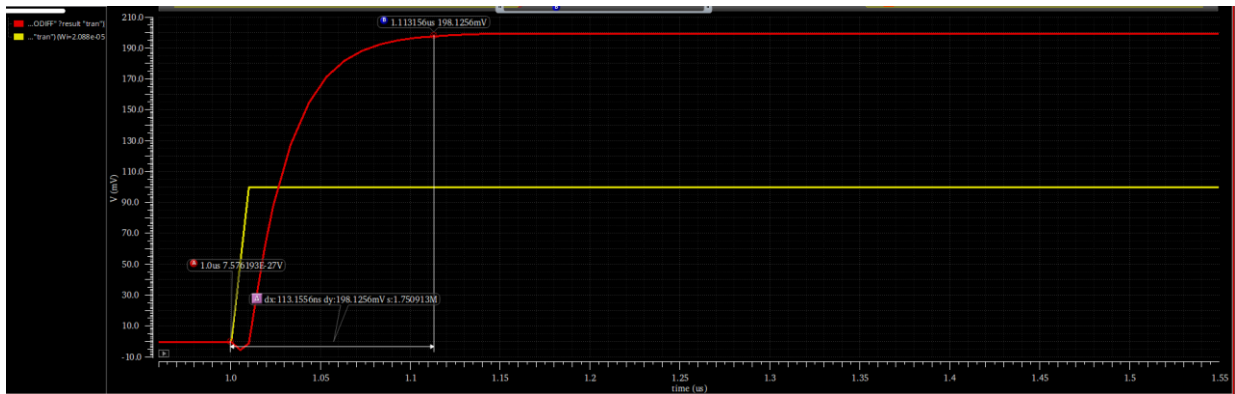


Figure 42: settling time before

To fix the settling time we should increase the GBW we can do that by increasing the width of the input pair to 30u instead of 21u and also increase the VCASCN to 1.3 instead of 1.2 and reduce VCASCP to 1 instead of 1.15 to increase the r_o thus increasing R_{out} .

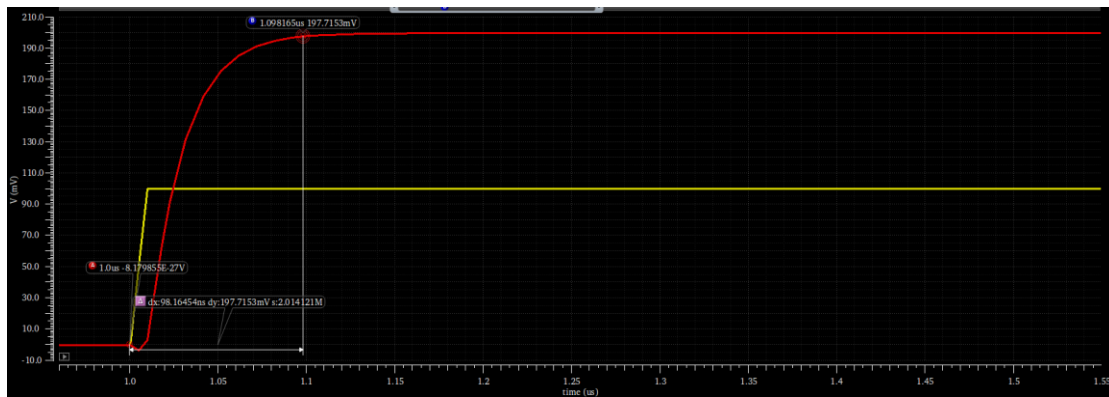


Figure 43: settling time after

Settling time = 98 ns which meets the specs.

6.2 Differential and CMFB loops stability (transient analysis):

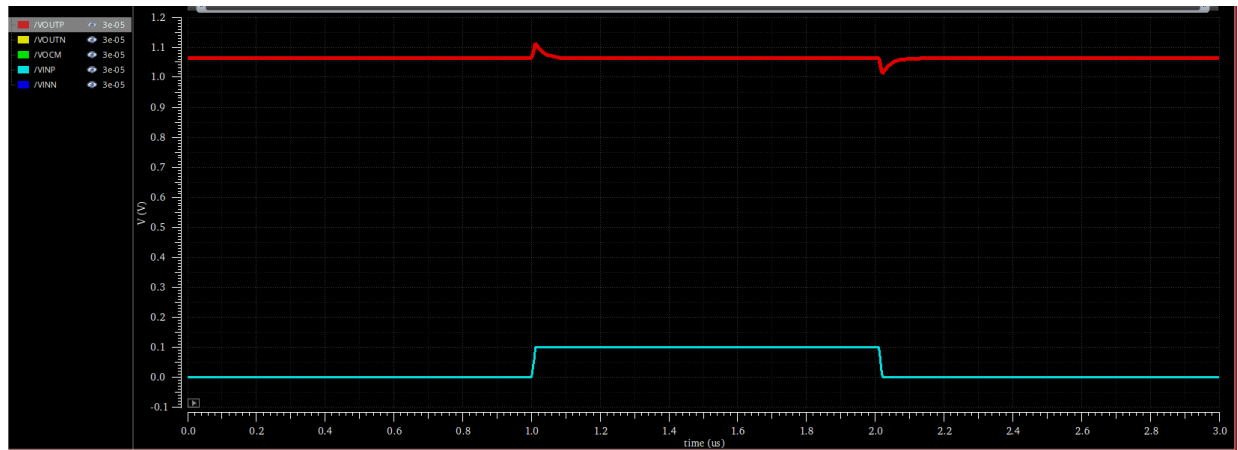


Figure 44: Transient response to a 100mV common-mode step pulse

No ringing in CM or DIFF there is a little overshoot due to the capacitors where in transitions which is high frequency the caps become short circuit this only in transition which is high freq.

6.3 Output swing:

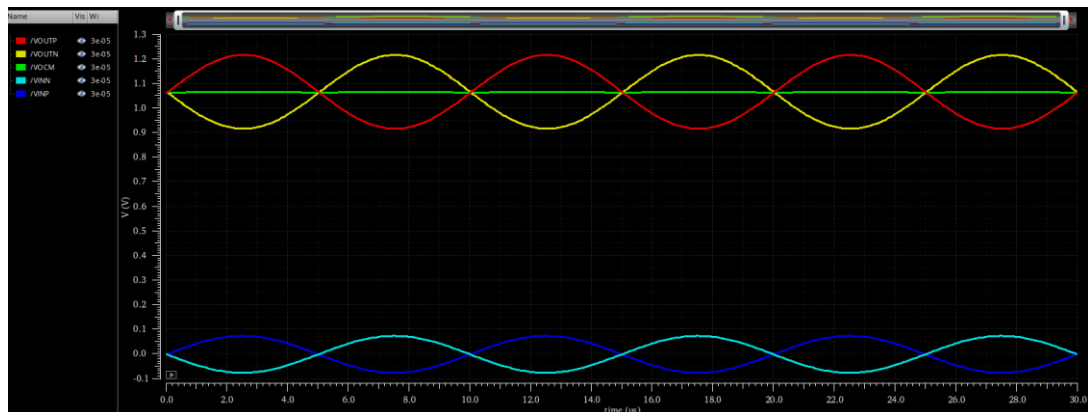


Figure 45: Transient output swing over three complete periods.

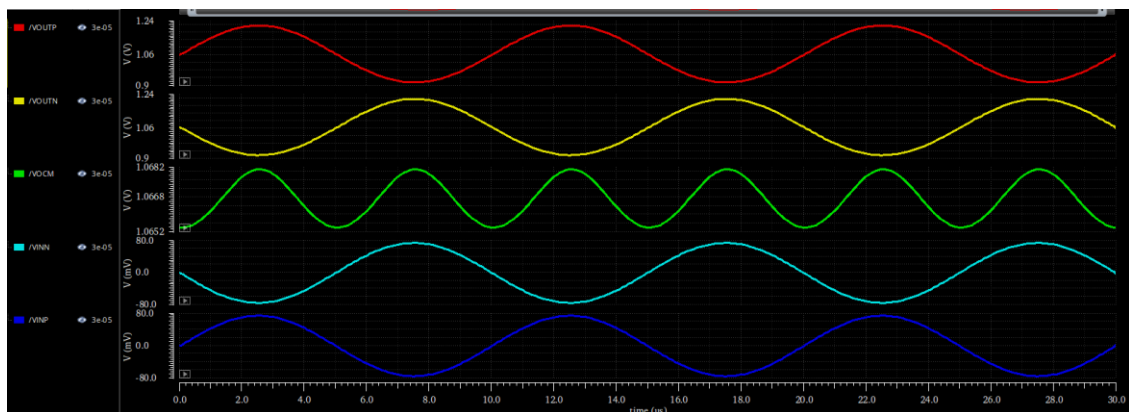


Figure 46: Zoomed-in view of the sinusoidal input and output waveforms.

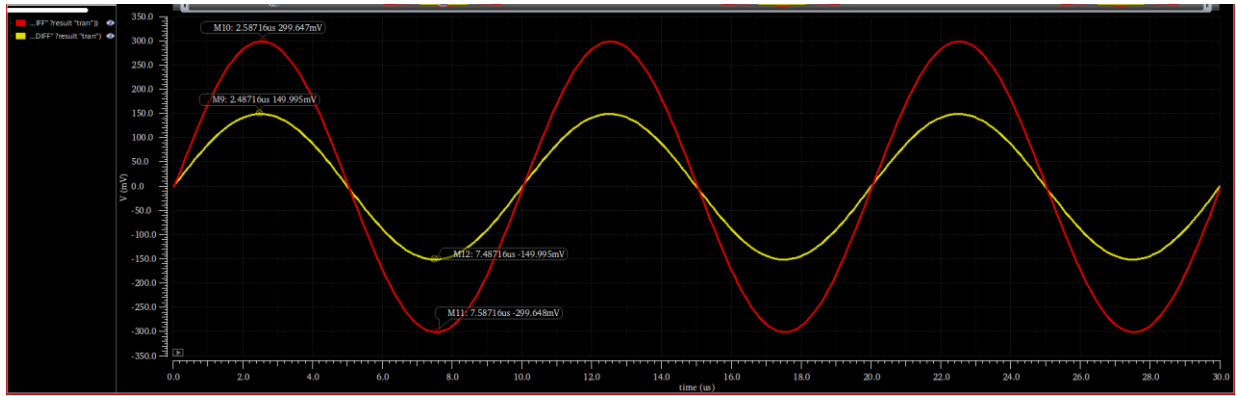


Figure 47: VIDIFF and VODIFF overlaid

As we see when the V_{in} amplitude = 150m the V_{out} is 300m that's because the closed loop gain = 2

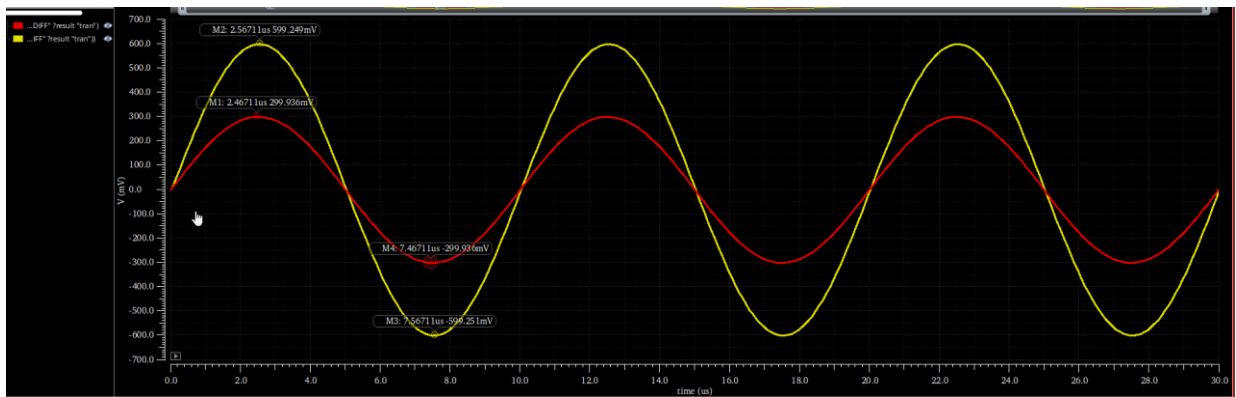


Figure 48: Vout peak to peak

As we see

$$V_{in_pp} = 600m$$

$$V_{out_pp} = 1.2V$$

$$So\ Av = V_{out_pp}/V_{in_pp} = 2$$

I did rough trans analysis by putting the amplitude of input = 1.1 to know what can my circuit do

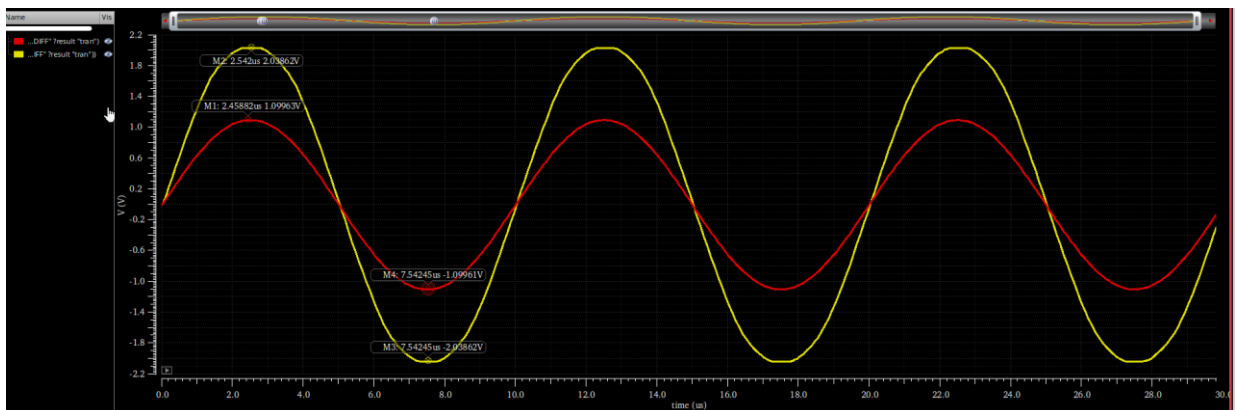


Figure 49: MAX peak to peak

As we see the circuit clipped at 2V which is 4V peak to peak.

6.3.1 CMIR Range:

I swept over V_{cm} and plotted GBW to know the CMIR

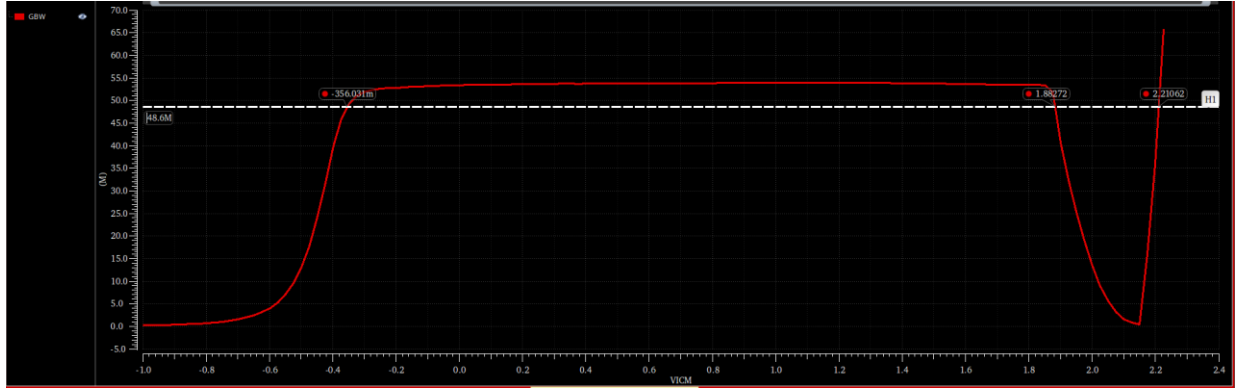


Figure 50: CMIR

$V_{in_low} = -350\text{m}$

$V_{in_high} = 1.9$

Which meets the specs !

7 Specs Achieved

Specification	Target Value	Achieved Result	Status
Supply Voltage (V_{DD})	2.5 V	2.5 V	Achieved
Closed-Loop Gain (A_{CL})	2 V/V	1.997 V/V	Achieved
Phase Margin (Differential)	$\geq 70^\circ$	89.44°	Achieved
Phase Margin (Common-Mode)	$\geq 70^\circ$	85.06°	Achieved
CM Input Range - Low	≤ 0 V	-350 mV	Achieved
CM Input Range - High	≥ 1 V	1.9 V	Achieved
Differential Output Swing	$\geq 1.2 V_{pk-pk}$	1.656 V_{pk-pk} (4 max V_{pk-pk})	Achieved
Load Capacitance (C_L)	500 fF	500 fF	Achieved
DC Loop Gain	≥ 60 dB	61.65 dB	Achieved
1% Settling Time	≤ 100 ns	98 ns (Rising Edge)	Achieved

8 Bonus Part:

8.1 Bias Circuit:

We will use magic battery for VCASCN and VCASCP

In the left circuit (VCASCN) we will sweep over W or L to get $V_{gs} = VCASCN = 1.2$ and the same for the right circuit we will sweep till we get $VCASCP = 1.15$

Also the diode connected devices is the same sizing the cascode devices we just sweep over the device we get VGS from.

I choose L and swept over W

NMOS: $L = 10\mu$

PMOS: $L = 2\mu$

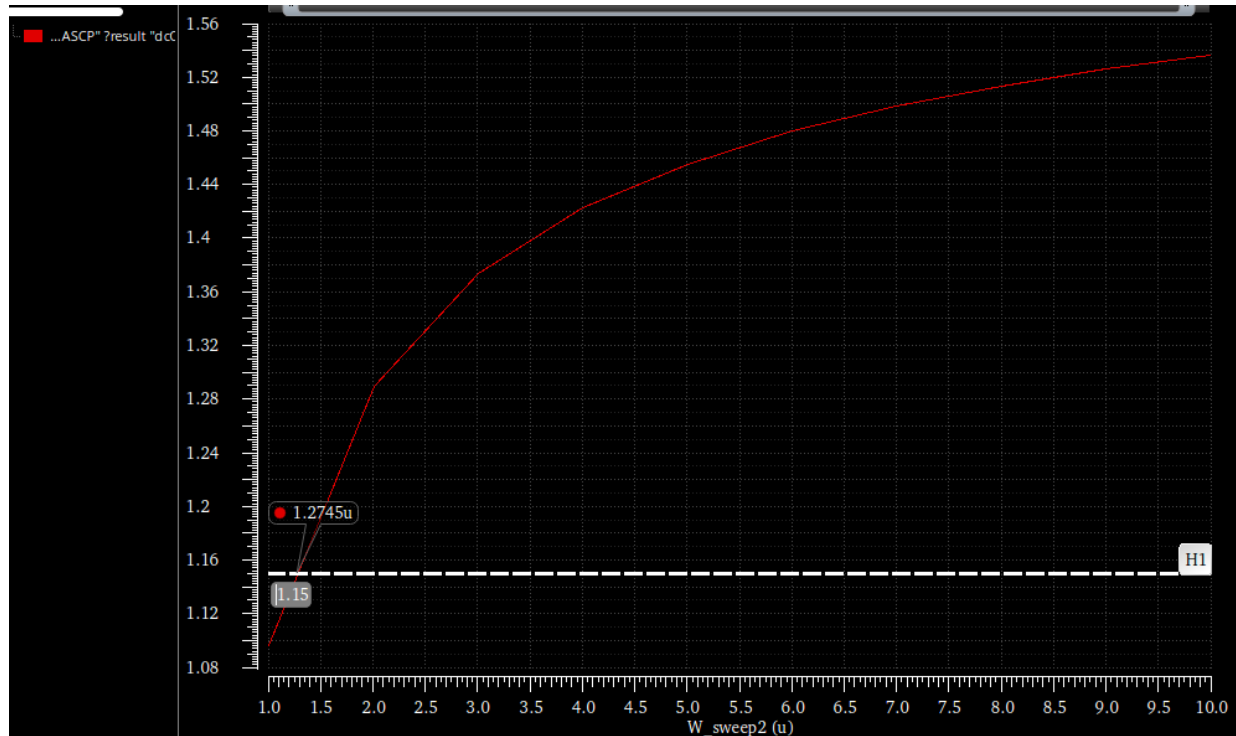
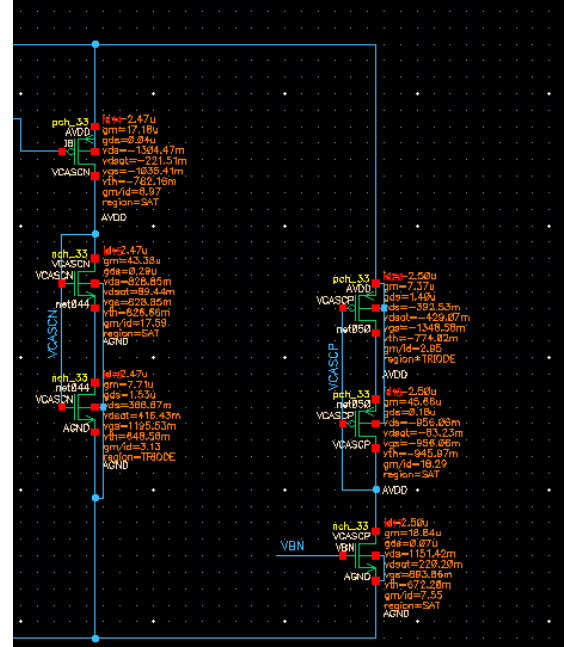


Figure 51: VCASCP

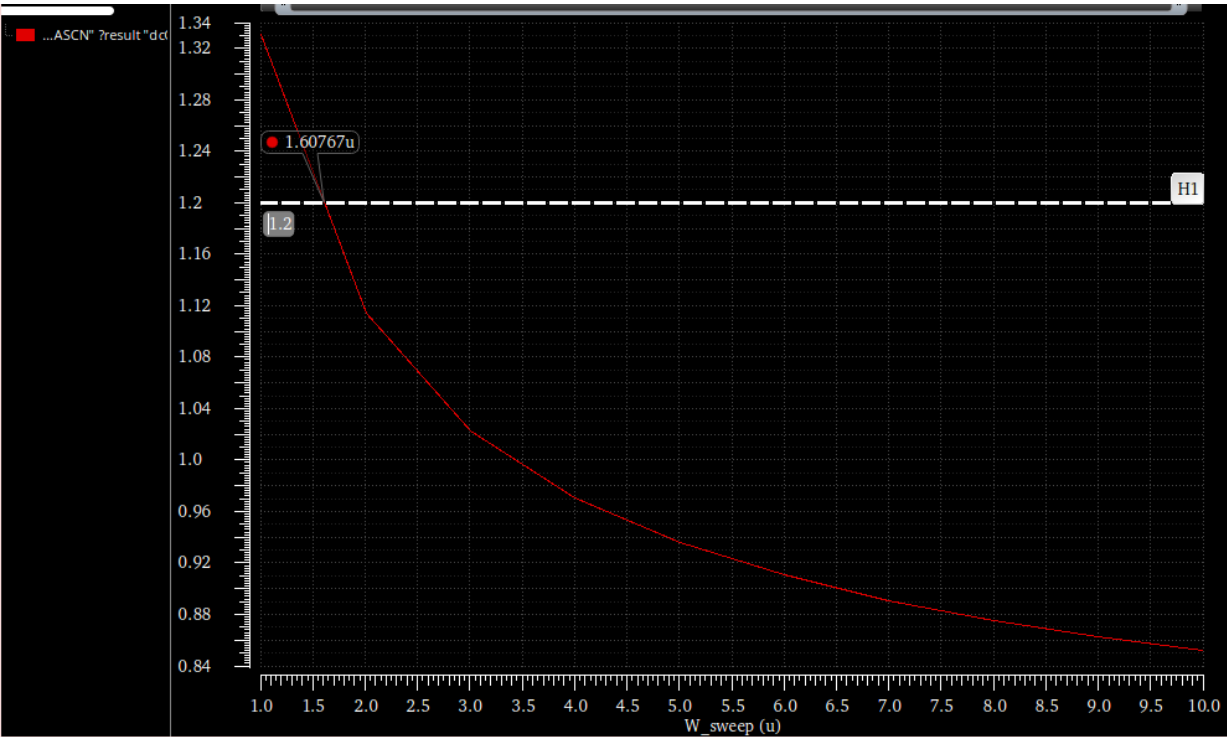


Figure 52: VCASCN

☒	☒	Ln	0.5u
☒	☒	Lp	0.5u
☒	☒	Wn	3.56u
☒	☒	Wp	15.3u
☒	☒	W	1.2u
☒	☒	Wi	21u
☒	☒	L_sweep	10u
☒	☒	L_sweep2	2u
☒	☒	W_sweep	1.5u
☒	☒	W_sweep2	1.2u

So this is the final sizing of the magic battery.

i designed the 2 magic batteries so that the total current of them equals 5u, as we see $I=4.97\mu$.

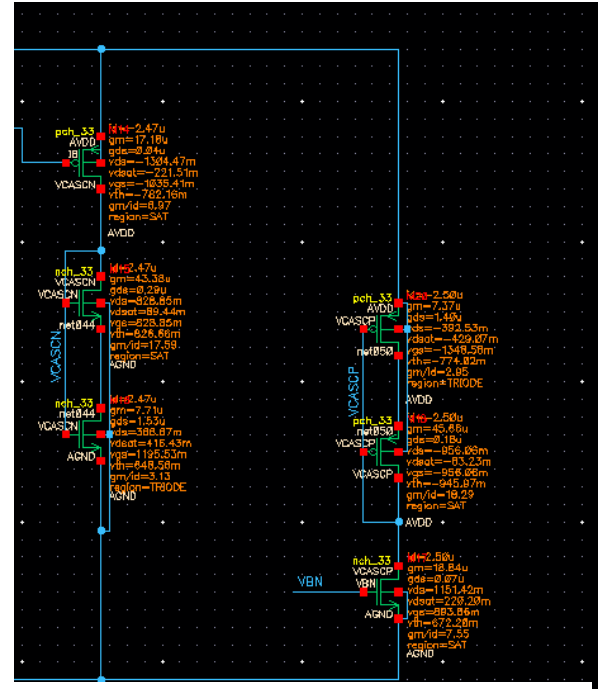


Figure 53: Magic battery OP

ITI_LABS:Folded_tb_d:1	v("/VCASCN" ?result "dcOp")	1.199
ITI_LABS:Folded_tb_d:1	v("/VCASCP" ?result "dcOp")	1.148

Here is the final bias voltages we got.

8.1.1 AC analysis :

I run Ac analysis for closed loop using the bias voltages we got

ITI_LABS:Folded_tb_d:1	VF("/VODIFF")				
ITI_LABS:Folded_tb_d:1	mag(VF("/VODIFF"))				
ITI_LABS:Folded_tb_d:1	VODIFF_dB				
ITI_LABS:Folded_tb_d:1	VODIFF_phase				
ITI_LABS:Folded_tb_d:1	Ao	1.998			
ITI_LABS:Folded_tb_d:1	Ao_dB	6.013			
ITI_LABS:Folded_tb_d:1	(Ao * BW)	16.53M			
ITI_LABS:Folded_tb_d:1	BW	8.27M			
ITI_LABS:Folded_tb_d:1	GBW	16.37M	> 15.3M		pass
ITI_LABS:Folded_tb_d:1	UGF	15.66M	> 15.3M		pass
ITI_LABS:Folded_tb_d:1	PM	83.1	> 70		pass

Which is the same results when we used a biasing voltage dc.

9 FOM

9.1 Total Current

9.1.1 OTA

Tail Source (M2): $20.82\mu\text{A}$

Top Cascode Sources (M3 + M4): $9.70 \mu\text{A} + 9.70 \mu\text{A} = 19.40 \mu\text{A}$

So $I_{\text{tot}} = 40 \mu\text{A}$

9.1.2 CMFB

CD Buffers (M1 + M2): $2.43 * 2 \mu\text{A} = 4.86 \mu\text{A}$

Error Amp Tail: $4.73 \mu\text{A}$

CMFB Subtotal: **$9.59 \mu\text{A}$**

Total System Current (I_{tot}): $40 \mu\text{A} + 9.59 \mu\text{A} = \mathbf{49.6 \mu\text{A}}$

9.2 AREA

9.2.1 OTA

Input Pair (M0, M1): $2 \times (20.88 \mu\text{m} \times 0.3 \mu\text{m}) = 12.53 \mu\text{m}^2$

Tail Source (M2): $1 \times (17.68 \mu\text{m} \times 1.0 \mu\text{m}) = 17.68 \mu\text{m}^2$

Top Sources (M3, M4): $2 \times (8.84 \mu\text{m} \times 1.0 \mu\text{m}) = 17.68 \mu\text{m}^2$

Top Cascodes (M5, M6): $2 \times (10 \mu\text{m} \times 0.5 \mu\text{m}) = 10 \mu\text{m}^2$

Bottom Sinks (M9, M10): $2 \times (5.52 \mu\text{m} \times 1.0 \mu\text{m}) = 11.04 \mu\text{m}^2$

Bottom Cascodes (M12, M13): $2 \times (2.5 \mu\text{m} \times 0.5 \mu\text{m}) = 2.5 \mu\text{m}^2$

Total Core OTA Active Area: $71.43 \mu\text{m}^2$

9.2.2 CMFB

For the diff

Input Pair (M0, M1): $2 \times (1.2 \mu\text{m} \times 1 \mu\text{m}) = 1.2 \mu\text{m}^2$

Tail Source (M2): $1 \times (1.38 \mu\text{m} \times 1.0 \mu\text{m}) = 1.38 \mu\text{m}^2$

Top Sources (M3, M4): $2 \times (2.21 \mu\text{m} \times 1.0 \mu\text{m}) = 4.42 \mu\text{m}^2$

For the sensing

CD: $2 \times (6.2 \mu\text{m} \times 0.5 \mu\text{m}) = 6.2 \mu\text{m}^2$

Top CM : $2 \times (2.21 \mu\text{m} \times 1.0 \mu\text{m}) = 4.42 \mu\text{m}^2$

Total CMFB Active Area: $17.62 \mu\text{m}^2$

TOTAL AREA = $89 \mu\text{m}^2$

$$FoM = \frac{UGF}{I_{tot} * AREA} = \frac{15.7 \text{ MHz}}{89\mu * 49.6\mu} = 3.6$$

That is very low to enhance it we will redesign the whole circuit so that the split ratio is not 1

We will set $S = 3.5$.